M.2 slots: NVMe and accelerator

slots_pcie.kicad_sch

M.2 slots: Wi-Fi and 5G modem, SIM

slots_radio.kicad_sch

Processor: PCIe and combo PHYs

soc_pcie.kicad_sch

Processor: USB and Type-C PHYs

soc_usb.kicad_sch

Processor: core supplies and ground

soc_power.kicad_sch

2.5GbE port 1: RTL8125BG

eth1.kicad_sch

2.5GbE port 2: RTL8125BG

eth2.kicad_sch

Power chip: RK806-1

pmic.kicad_sch

Core regulators: big cores, NPU, 1.1 V

core_regs.kicad_sch

Input converters: 5 V, 4 V and three 3.3 V rails

power_input.kicad_sch

Memory channel 0: LPDDR4X

ddr0.kicad_sch

Memory channel 1: LPDDR4X

ddr1.kicad_sch

Processor: clock, reset, IO supplies, GPIO bank 0

soc_system.kicad_sch

Processor: GPIO banks 1 and 2

soc_gpio12.kicad_sch

Processor: GPIO banks 3 and 4

soc_gpio34.kicad_sch

PCIe reference clock: oscillator and two buffers

pcie_clock.kicad_sch

Processor: display and camera blocks (unused)

USB-C port 1: USB 3, DisplayPort, power in

USB-C port 2: USB 3, DisplayPort, power in

USB 2.0 hub and USB-A ports

HookProbe

Sheet: /

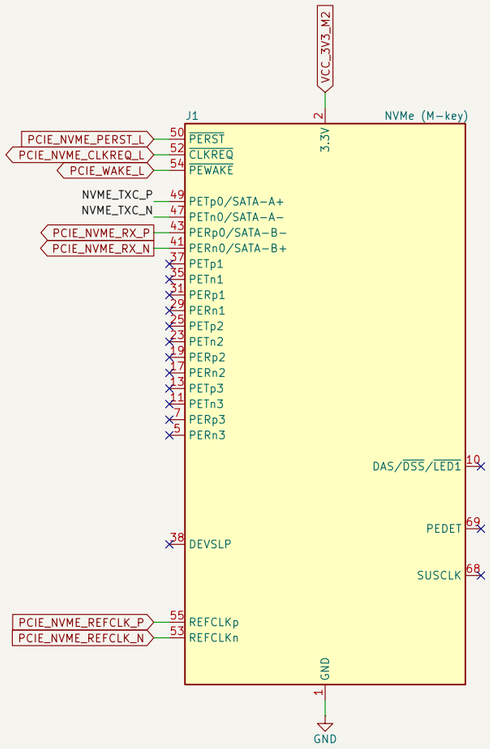
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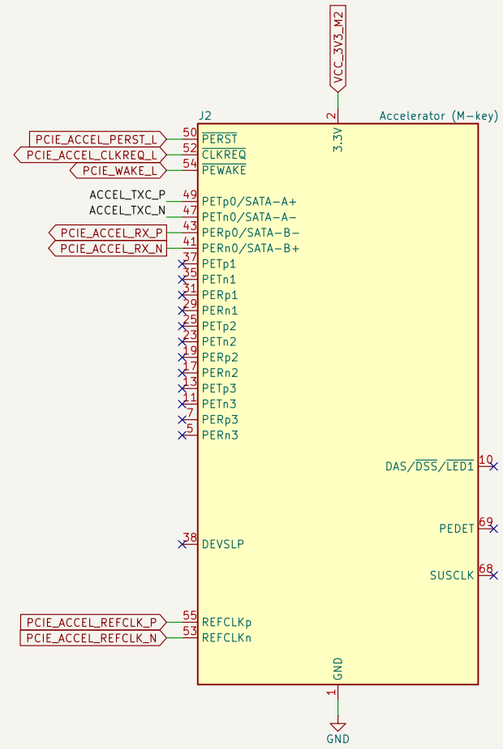
Title: HookProbe SBC rev A

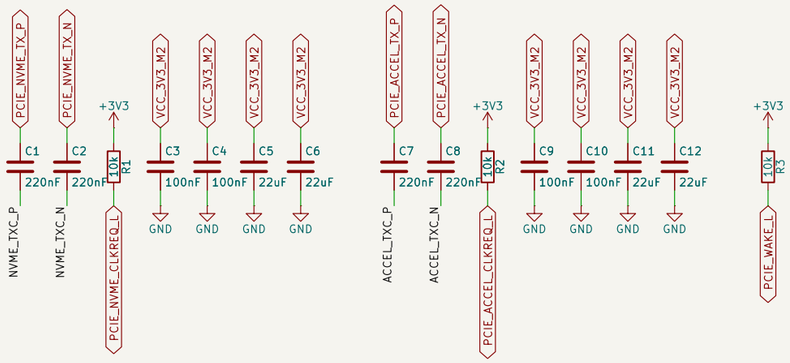
Size: A4 | Date: | **Rev: A0**

KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 | Id: 1/25

M.2 M-key slots, one PCIe Gen3 lane each.
J1 NVMe SSD: RK3588 controller pcie3x4, PCIe3 PHY lane 0.
J2 NPU / FPGA card: RK3588 controller pcie3x2, PCIe3 PHY lane 1.
100 MHz reference clocks come from the PCIe clock buffer (sheet not drawn yet).
PCIE_WAKE_L is shared by all PCIe slots (open drain, one pull-up here).







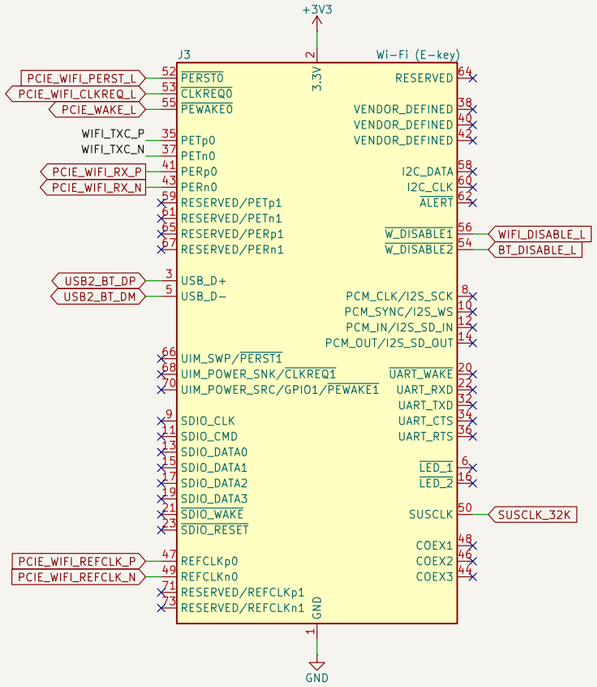
HookProbe

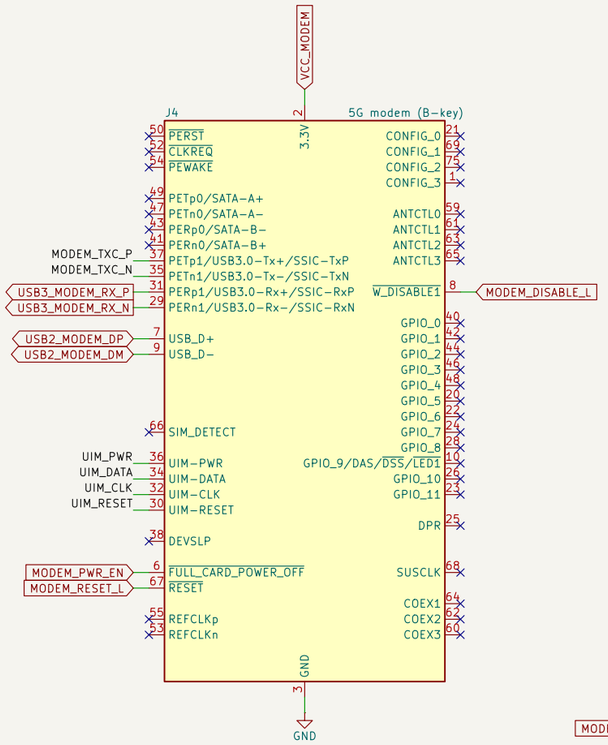
Sheet: /M.2 slots: NVMe and accelerator/
File: slots_pcie.kicad_sch

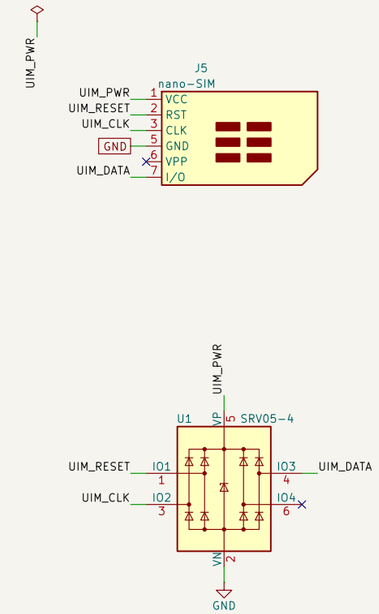
Title: M.2 slots: NVMe and accelerator

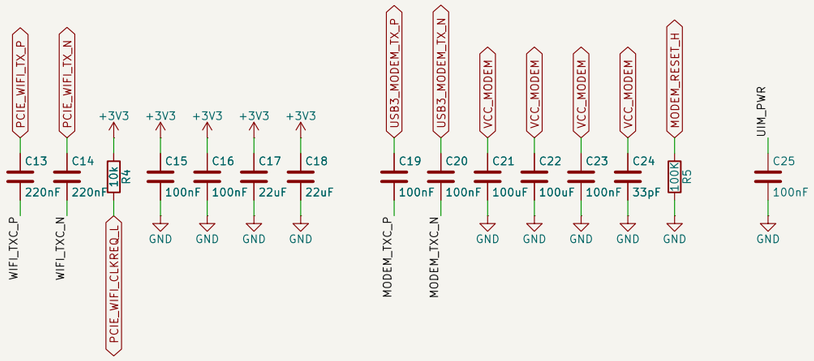
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 2/25

J3 Wi-Fi (E-key): PCIe on RK3588 pcie2x1l2 / combo PHY 0, Bluetooth on USB 2.
J4 5G modem (B-key): USB 3 + USB 2 from the hub, SIM on J5.
VCC_MODEM is a dedicated 3.3 V rail sized for transmit bursts.
MODEM_PWR_EN high switches the modem on (card pulls it low by default).
MODEM_RESET_H high resets the modem (transistor pulls the card's 1.8 V reset pin low).
MODEM_VCC_EN switches the whole VCC_MODEM rail, for a hard power cycle.
SIM detect is not wired in rev A: the holder has no switch, so no SIM hot-swap.









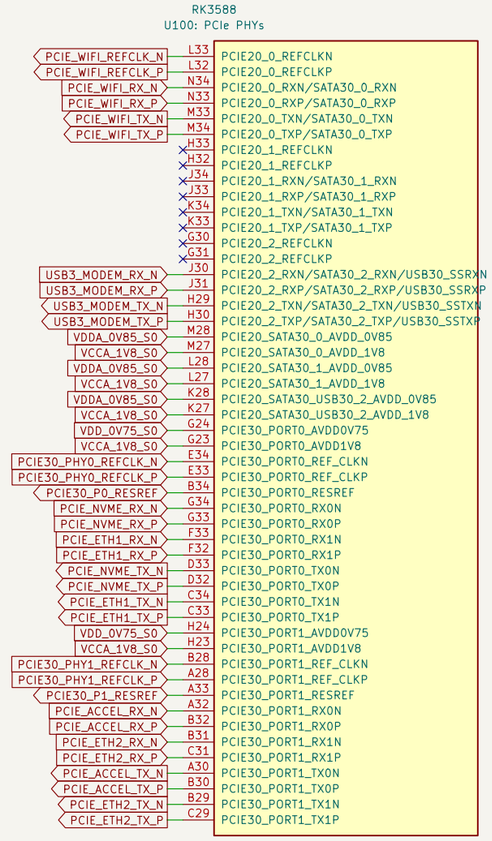
HookProbe

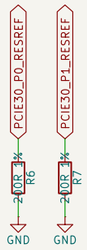
Sheet: /M.2 slots: Wi-Fi and 5G modem, SIM/
File: slots_radio.kicad_sch

Title: M.2 slots: Wi-Fi and 5G modem, SIM

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 3/25

RK3588 unit 1: PCIe3 PHY (4 lanes, split 4 x 1) and the three combo PHYs.
Lane to slot mapping is in PCIE3_LANES in build.py and is still to be confirmed
against Rockchip's PCIe guide. Combo PHY 1 is spare (usable for SATA only).
RESREF resistors are 200R 1%, as on the Radxa Rock 5B v1.45 schematic (sheet 18).
Supply pins go to rails named after the PHY; filtering and decoupling belong on the power sheet.





HookProbe
Sheet: /Processor: PCIe and combo PHYs/
File: soc_pcie.kicad_sch
Title: Processor: PCIe and combo PHYs
Size: A3 | Date: | **Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 | Id: 4/25

RK3588 unit 2. TYPEC0 feeds USB-C port 1, TYPEC1 feeds USB-C port 2 (USB 3 and DisplayPort share the same four pairs). USB20_HOST0 is the USB 2 side of the hub; USB20_HOST1 goes to the Wi-Fi card's Bluetooth. The OTG ID balls are unused (a Type-C port gets its role from the PD controller); VBUSDET is pulled up on the port sheets. REXT values follow the Radxa Rock 5B v1.45 schematic (sheet 14).

RK3588
U100: USB and Type-C PHYs

TYPEC0_DP0_REXT	AH16	TYPEC0_DP0_REXT
VDDA_0V85_S0	AH14	TYPEC0_DP0_VDDA_0V85
VCCA_1V8_S0	AG14	TYPEC0_DP0_VDDH_1V8
VDDA_0V85_S0	AJ14	TYPEC0_DP0_VDD_0V85
USBC1_SBU1	AL15	TYPEC0_SBU1/DP0_AUXP
USBC1_SBU2	AM15	TYPEC0_SBU2/DP0_AUXN
USBC1_SSRX1_N	AP13	TYPEC0_SSRX1N/DP0_TX0N
USBC1_SSRX1_P	AN13	TYPEC0_SSRX1P/DP0_TX0P
USBC1_SSRX2_N	AP15	TYPEC0_SSRX2N/DP0_TX2N
USBC1_SSRX2_P	AN15	TYPEC0_SSRX2P/DP0_TX2P
USBC1_SSTX1_N	AN14	TYPEC0_SSTX1N/DP0_TX1N
USBC1_SSTX1_P	AP14	TYPEC0_SSTX1P/DP0_TX1P
USBC1_SSTX2_N	AN16	TYPEC0_SSTX2N/DP0_TX3N
USBC1_SSTX2_P	AP16	TYPEC0_SSTX2P/DP0_TX3P
TYPEC0_USB20_OTG0_REXT	AP12	TYPEC0_USB20_OTG0_REXT
USBC1_USB2_DM	AM12	TYPEC0_USB20_OTG_DM
USBC1_USB2_DP	AL12	TYPEC0_USB20_OTG_DP
	AL14	TYPEC0_USB20_OTG_ID
	AM14	
USBC1_VBUSDET	AG16	TYPEC0_USB20_VBUSDET
TYPEC1_DP1_REXT	AJ13	TYPEC1_DP1_REXT
VDDA_0V85_S0	AG13	TYPEC1_DP1_VDDA_0V85
VCCA_1V8_S0	AH13	TYPEC1_DP1_VDDH_1V8
VDDA_0V85_S0	AL10	TYPEC1_DP1_VDD_0V85
USBC2_SBU1	AM10	TYPEC1_SBU1/DP1_AUXP
USBC2_SBU2	AP8	TYPEC1_SBU2/DP1_AUXN
USBC2_SSRX1_N	AN8	TYPEC1_SSRX1N/DP1_TX0N
USBC2_SSRX1_P	AP10	TYPEC1_SSRX1P/DP1_TX0P
USBC2_SSRX2_N	AN10	TYPEC1_SSRX2N/DP1_TX2N
USBC2_SSRX2_P	AN9	TYPEC1_SSRX2P/DP1_TX2P
USBC2_SSTX1_N	AP9	TYPEC1_SSTX1N/DP1_TX1N
USBC2_SSTX1_P	AN11	TYPEC1_SSTX1P/DP1_TX1P
USBC2_SSTX2_N	AP11	TYPEC1_SSTX2N/DP1_TX3N
USBC2_SSTX2_P	AP7	TYPEC1_SSTX2P/DP1_TX3P
TYPEC1_USB20_OTG1_REXT	AL9	TYPEC1_USB20_OTG1_REXT
USBC2_USB2_DM	AK9	TYPEC1_USB20_OTG_DM
USBC2_USB2_DP	AK8	TYPEC1_USB20_OTG_DP
	AL8	TYPEC1_USB20_OTG_ID
USBC2_VBUSDET	AG11	TYPEC1_USB20_VBUSDET
VCCA_1V8_S0	AJ10	USB20_AVDD_1V8
VCCA_3V3_S0	AH10	USB20_AVDD_3V3
VDDA_0V75_S0	AL6	USB20_DVDD_0V75
USB2_HUB_DM	AK6	USB20_HOST0_DM
USB2_HUB_DP	AG9	USB20_HOST0_DP
USB20_HOST0_REXT	AM7	USB20_HOST0_REXT
USB2_BT_DM	AL7	USB20_HOST1_DM
USB2_BT_DP	AH9	USB20_HOST1_DP
USB20_HOST1_REXT		USB20_HOST1_REXT

TYPEC0_DP0_REXT — 2K 1% R8 — GND
TYPEC0_USB20_OTG0_REXT — 200R 1% R9 — GND
TYPEC1_DP1_REXT — 2K 1% R10 — GND
TYPEC1_USB20_OTG1_REXT — 200R 1% R11 — GND
USB20_HOST0_REXT — 200R 1% R12 — GND
USB20_HOST1_REXT — 200R 1% R13 — GND

HookProbe
Sheet: /Processor: USB and Type-C PHYs/
File: soc_usb.kicad_sch
Title: Processor: USB and Type-C PHYs
Size: A3 | Date: | Rev: A0
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 | Id: 5/25

RK3588 units 11 and 13, with rail assignment and decoupling from the Radxa Rock 5B v1.45 schematic,
sheet 10. The MEM pin groups share their core rail, except CPU_BIG1_MEM which has its own rail.
Net ties join each remotely sensed rail to its _FB sense net: place them at the processor balls.
All 421 ground balls are stacked on the single VSS pin of the ground unit.

RK3588
U100: Core supplies

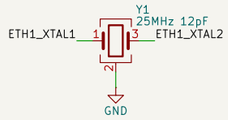
VDD_CPU_BIG0_S0	M16	VDD_CPU_BIG0_0
VDD_CPU_BIG0_S0	N16	VDD_CPU_BIG0_1
VDD_CPU_BIG0_S0	P16	VDD_CPU_BIG0_2
VDD_CPU_BIG0_S0	R16	VDD_CPU_BIG0_3
VDD_CPU_BIG0_S0	T16	VDD_CPU_BIG0_4
VDD_CPU_BIG0_S0	T17	VDD_CPU_BIG0_5
VDD_CPU_BIG0_S0	R17	VDD_CPU_BIG0_6
VDD_CPU_BIG0_S0	P17	VDD_CPU_BIG0_7
VDD_CPU_BIG0_S0	N17	VDD_CPU_BIG0_8
VDD_CPU_BIG0_S0	M17	VDD_CPU_BIG0_9
VDD_CPU_BIG0_S0	M19	VDD_CPU_BIG0_MEM_0
VDD_CPU_BIG0_S0	N19	VDD_CPU_BIG0_MEM_1
VDD_CPU_BIG1_S0	K24	VDD_CPU_BIG1_0
VDD_CPU_BIG1_S0	L24	VDD_CPU_BIG1_1
VDD_CPU_BIG1_S0	M24	VDD_CPU_BIG1_2
VDD_CPU_BIG1_S0	N24	VDD_CPU_BIG1_3
VDD_CPU_BIG1_S0	P24	VDD_CPU_BIG1_4
VDD_CPU_BIG1_S0	P23	VDD_CPU_BIG1_5
VDD_CPU_BIG1_S0	N23	VDD_CPU_BIG1_6
VDD_CPU_BIG1_S0	M23	VDD_CPU_BIG1_7
VDD_CPU_BIG1_S0	L23	VDD_CPU_BIG1_8
VDD_CPU_BIG1_S0	K23	VDD_CPU_BIG1_9
VDD_CPU_BIG1_MEM_S0	M21	VDD_CPU_BIG1_MEM_0
VDD_CPU_BIG1_MEM_S0	N21	VDD_CPU_BIG1_MEM_1
VDD_CPU_LIT_S0	U22	VDD_CPU_LIT_0
VDD_CPU_LIT_S0	V22	VDD_CPU_LIT_1
VDD_CPU_LIT_S0	W22	VDD_CPU_LIT_2
VDD_CPU_LIT_S0	Y22	VDD_CPU_LIT_3
VDD_CPU_LIT_S0	Y21	VDD_CPU_LIT_4
VDD_CPU_LIT_S0	W21	VDD_CPU_LIT_5
VDD_CPU_LIT_S0	V21	VDD_CPU_LIT_6
VDD_CPU_LIT_S0	U21	VDD_CPU_LIT_7
VDD_CPU_LIT_S0	T22	VDD_CPU_LIT_MEM_0
VDD_CPU_LIT_S0	T21	VDD_CPU_LIT_MEM_1
VDD_GPU_S0	AA13	VDD_GPU_0
VDD_GPU_S0	AB13	VDD_GPU_1
VDD_GPU_S0	AC13	VDD_GPU_2
VDD_GPU_S0	AD13	VDD_GPU_3
VDD_GPU_S0	AD14	VDD_GPU_4
VDD_GPU_S0	AC14	VDD_GPU_5
VDD_GPU_S0	AB14	VDD_GPU_6
VDD_GPU_S0	AA14	VDD_GPU_7
VDD_GPU_S0	AD15	VDD_GPU_8
VDD_GPU_S0	AC15	VDD_GPU_9
VDD_GPU_S0	AB15	VDD_GPU_10
VDD_GPU_S0	AA15	VDD_GPU_11
VDD_GPU_S0	AA12	VDD_GPU_MEM_0
VDD_GPU_S0	AB12	VDD_GPU_MEM_1
VDD_LOG_S0	AD17	VDD_LOGIC_0
VDD_LOG_S0	AD18	VDD_LOGIC_1
VDD_LOG_S0	AD19	VDD_LOGIC_2
VDD_LOG_S0	AC19	VDD_LOGIC_3
VDD_LOG_S0	AC18	VDD_LOGIC_4
VDD_LOG_S0	AC17	VDD_LOGIC_5
VDD_LOG_S0	V16	VDD_LOGIC_6
VDD_LOG_S0	V17	VDD_LOGIC_7
VDD_LOG_S0	K19	VDD_LOGIC_8
VDD_LOG_S0	K20	VDD_LOGIC_9
VDD_NPU_S0	AD23	VDD_NPU_0
VDD_NPU_S0	AC23	VDD_NPU_1
VDD_NPU_S0	AB23	VDD_NPU_2
VDD_NPU_S0	AD22	VDD_NPU_3
VDD_NPU_S0	AC22	VDD_NPU_4
VDD_NPU_S0	AB22	VDD_NPU_5
VDD_NPU_S0	AB21	VDD_NPU_6
VDD_NPU_S0	AE22	VDD_NPU_MEM_0
VDD_NPU_S0	AE23	VDD_NPU_MEM_1
VDD_VDENC_S0	R14	VDD_VDENC_0
VDD_VDENC_S0	T14	VDD_VDENC_1
VDD_VDENC_S0	U14	VDD_VDENC_2
VDD_VDENC_S0	V14	VDD_VDENC_3
VDD_VDENC_S0	W14	VDD_VDENC_4
VDD_VDENC_S0	W13	VDD_VDENC_5
VDD_VDENC_S0	V12	VDD_VDENC_MEM_0
VDD_VDENC_S0	V13	VDD_VDENC_MEM_1

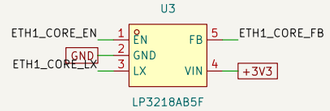
RK3588
U100: Ground

GND	H28	VSS

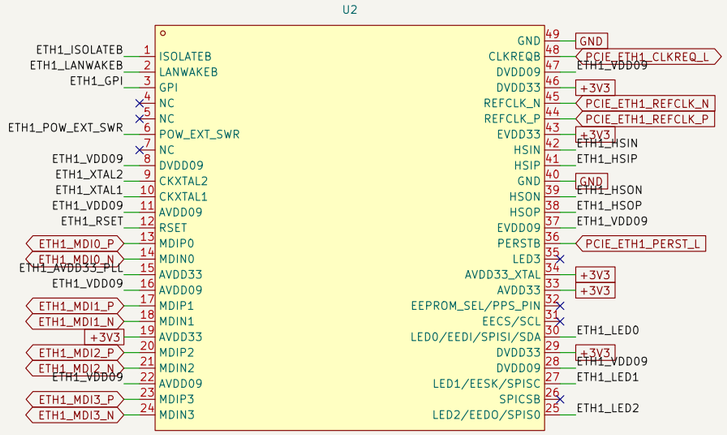
| | | |

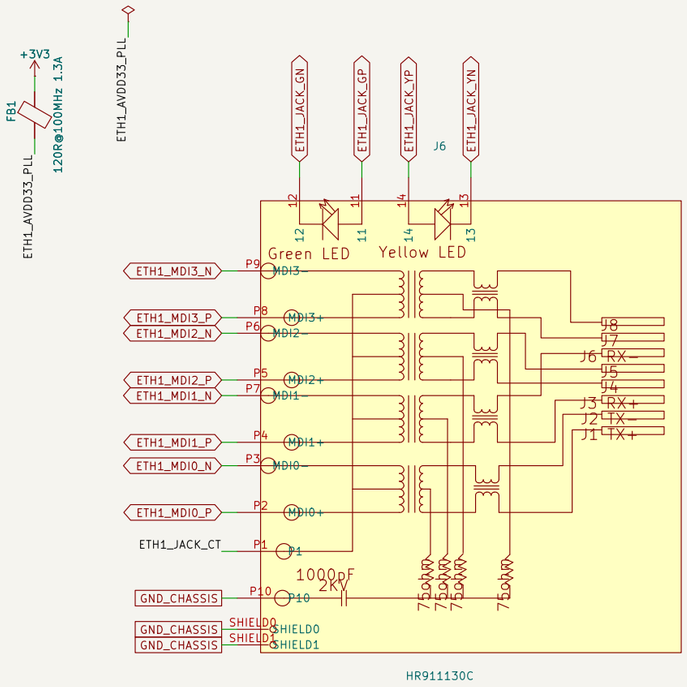
Copied from the Radxa Rock 5B v1.45 schematic, sheet 24 (RTL8125BG, external 0.95 V core supply).
Left out on purpose: the zero-ohm option links, the unpopulated strap resistors, and the
common-mode/ESD filters on the MDI pairs (the reference part could not be matched to an LCSC part).
Jack: HanRun HR911130C, pinout taken from its datasheet (vendor/lcsc_C50933/datasheet.pdf), which
matches the LCSC symbol and not the Radxa drawing. It is rated 100/1000Base-T only: validate it at
2.5 Gb or substitute a 2.5G-rated jack. Centre tap P1 has a capacitor to ground only, as in the reference.
Reference clock comes from the PCIe clock buffer (sheet not drawn yet).

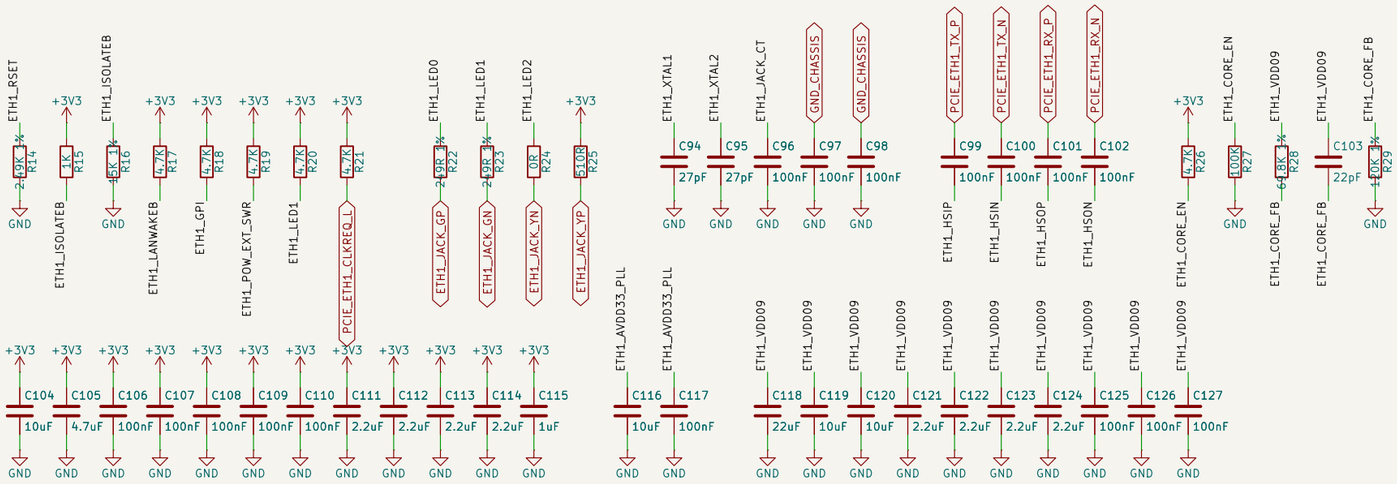












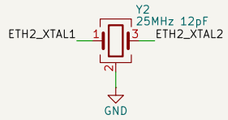
HookProbe

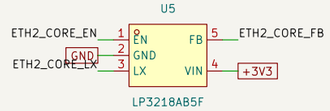
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File: eth1.kicad_sch

Title: 2.5GbE port 1: RTL8125BG

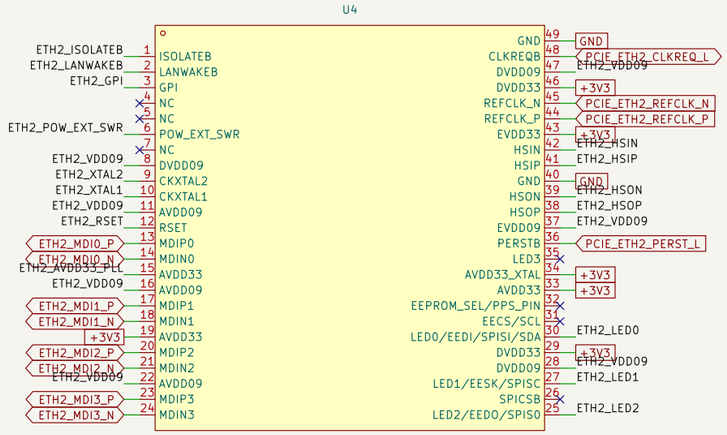
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 7/25

Copied from the Radxa Rock 5B v1.45 schematic, sheet 24 (RTL8125BG, external 0.95 V core supply).
Left out on purpose: the zero-ohm option links, the unpopulated strap resistors, and the
common-mode/ESD filters on the MDI pairs (the reference part could not be matched to an LCSC part).
Jack: HanRun HR911130C, pinout taken from its datasheet (vendor/lcsc_C50933/datasheet.pdf), which
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2.5 Gb or substitute a 2.5G-rated jack. Centre tap P1 has a capacitor to ground only, as in the reference.
Reference clock comes from the PCIe clock buffer (sheet not drawn yet).

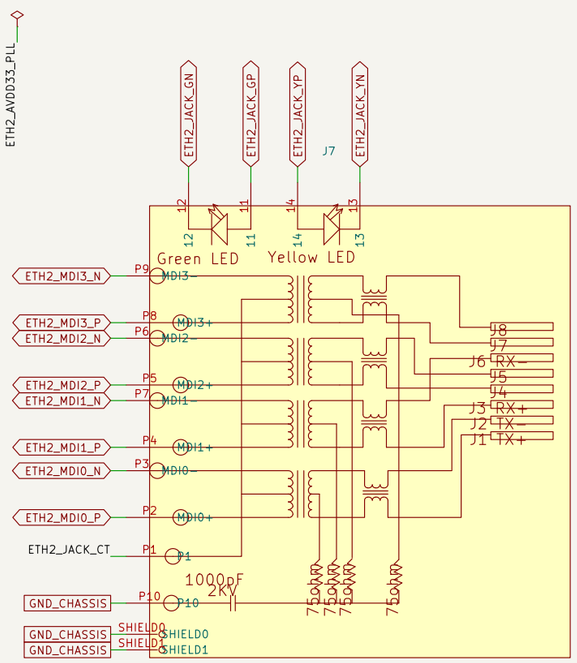


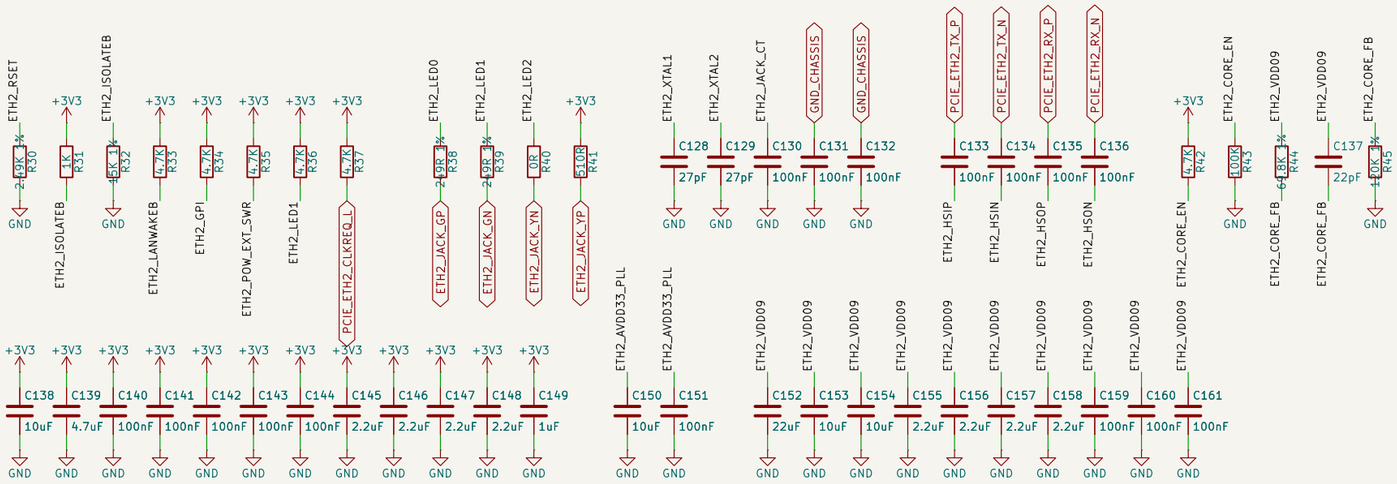








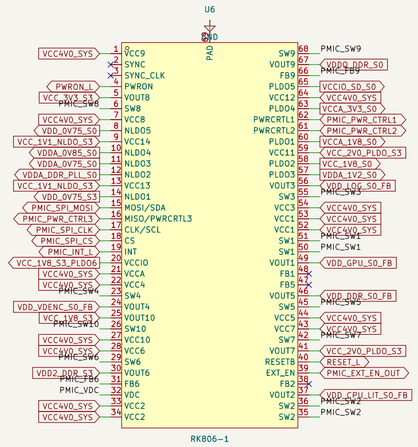


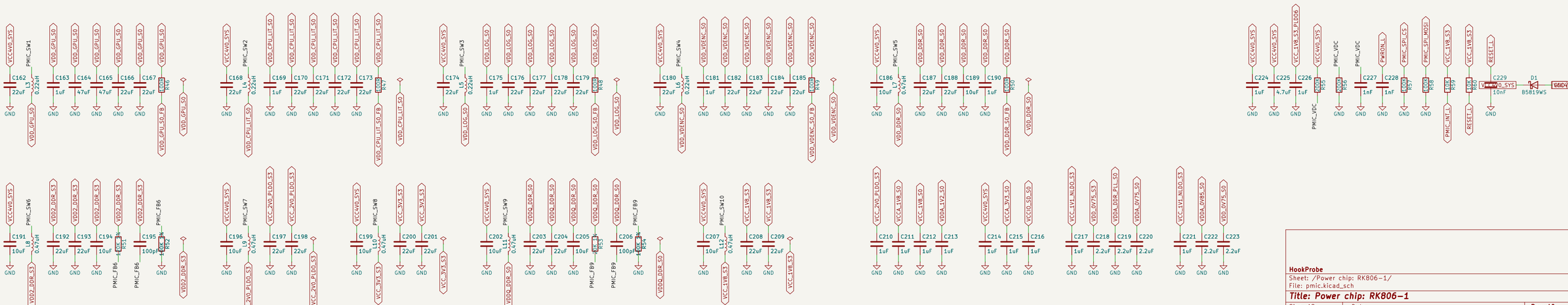


HookProbe

Sheet: /2.5GbE port 2: RTL8125BG/		
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Title: 2.5GbE port 2: RTL8125BG		
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 8/25

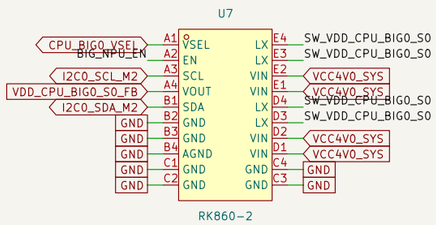
Copied from the Radxa Rock 5B v1.45 schematic, sheet 22. Rail names are the reference's.
VCC4V0_SYS (4.0 V system rail) and VCC_1V1_NLDO_S3 come from regulators that are not drawn yet.
Bucks 1–5 are sensed at the processor: each VOUT pin goes to a <rail>_FB net with 100R to the local
output; the sense net must be tied to the rail at the processor balls (net tie, not drawn yet).
Buck 6 and 9 dividers are the LPDDR4/4X values (1.1 V and 0.6 V); LPDDR5 needs 110K and a OR/open pair.
J (3-pin header): short pin 1 to 2 for the power button, pin 3 to 2 for reset.

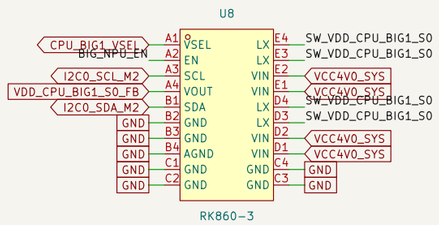


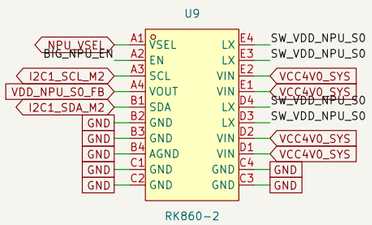


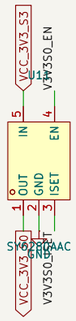
HookProbe
Sheet: /Power chip: RK806-1/
File: pmic.kicad_sch
Title: Power chip: RK806-1
Size: A2 Date:
Rev: A0
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 Id: 9/25

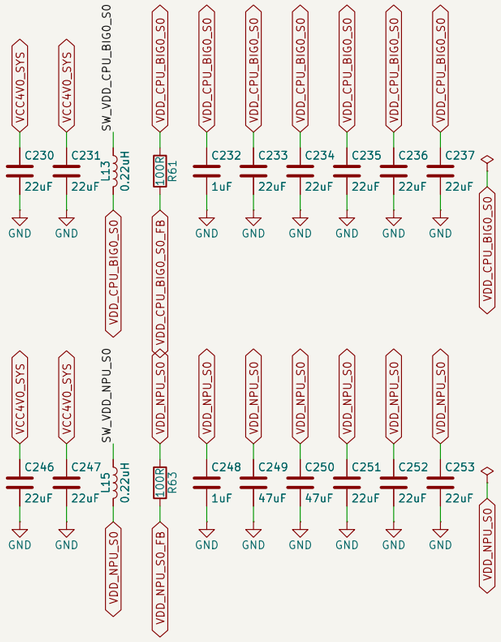
Copied from the Radxa Rock 5B v1.45 schematic, sheet 23.
U (RK860-2) big core 0 and U (RK860-3) big core 1 share I2C0; the NPU regulator (RK860-2) is on I2C1.
Each VOUT pin is the remote sense input: it goes to <rail>_FB, tied to the rail at the processor.
VCC_3V3_S0 switch: SY6280AAC in place of the reference's SGM2576/TT9107 (same pinout and current-limit formula, 6800 / R in amps; 4.7K gives 1.45 A). The reference's LCD supply is not needed.

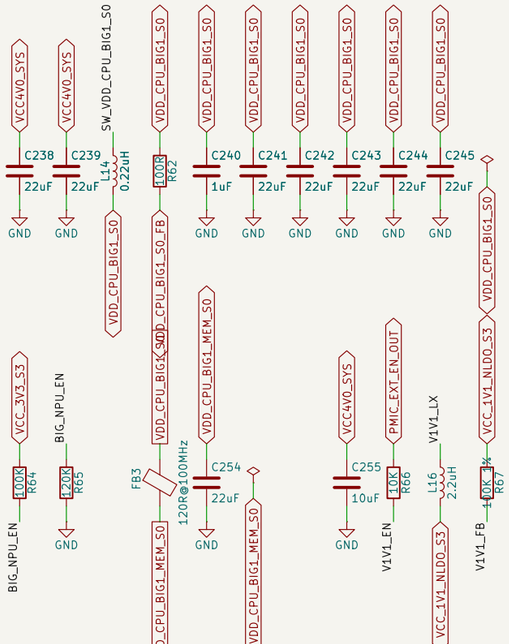


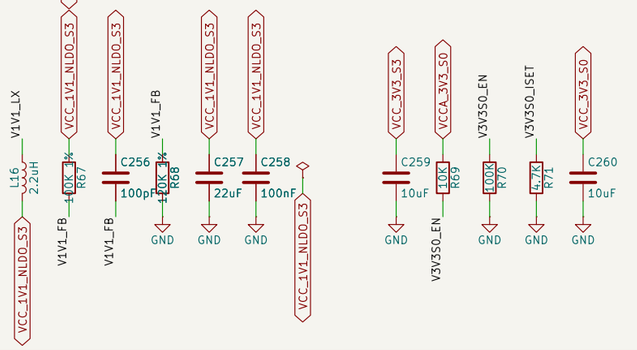












HookProbe

Sheet: /Core regulators: big cores, NPU, 1.1 V/
File: core_regs.kicad_sch

Title: Core regulators: big cores, NPU, 1.1 V

Size: A3	Date:	Rev: A0
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 10/25

Five MP8759 converters, each copied from the 5 V stage of the Radxa Rock 5B v1.45 schematic, sheet 20.
Differences from the reference, to be checked against the MP8759 datasheet before layout:
- all five run directly from VIN (the reference feeds its 4 V stage from the 5 V rail);
- the three 3.3 V rails are new, with a 47K / 10.2K divider (0.6 V reference gives 3.36 V);
- every stage uses 1.5 uH, the datasheet's table value for 3.3 V and 5 V outputs;
- ENABLE: the pin's absolute maximum is 4.5 V. The 5 V and 4 V stages start with VIN through a 100K / 75K divider clamped by a 3.6 V zener. The three 3.3 V stages are switched by logic: +3V3 and VCC_3V3_M2 follow VCC_3V3_S0, so nothing feeds the processor's pins before its own supplies are up; VCC_MODEM is switched by software so a hung modem can be power-cycled.

26 V is the MP8759's ABSOLUTE MAXIMUM input, not a working limit: power the board from 20 V or less (USB-C PD up to 20 V, or a 12 V to 19 V DC supply). The power entry sheet feeds VIN.

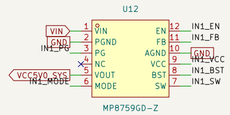
5.05 V: USB ports (5.25 V is the limit with tolerances). On whenever VIN is present

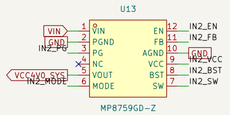
4.04 V: power chip and core regulators. On whenever VIN is present

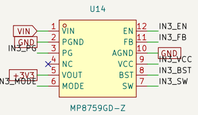
3.36 V: Wi-Fi, Ethernet, logic. Follows VCC_3V3_S0

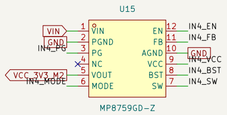
3.36 V: NVMe and accelerator slots. Follows VCC_3V3_S0

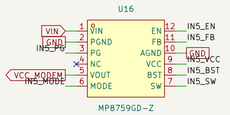
3.36 V: 5G modem. Switched by software (MODEM_VCC_EN)

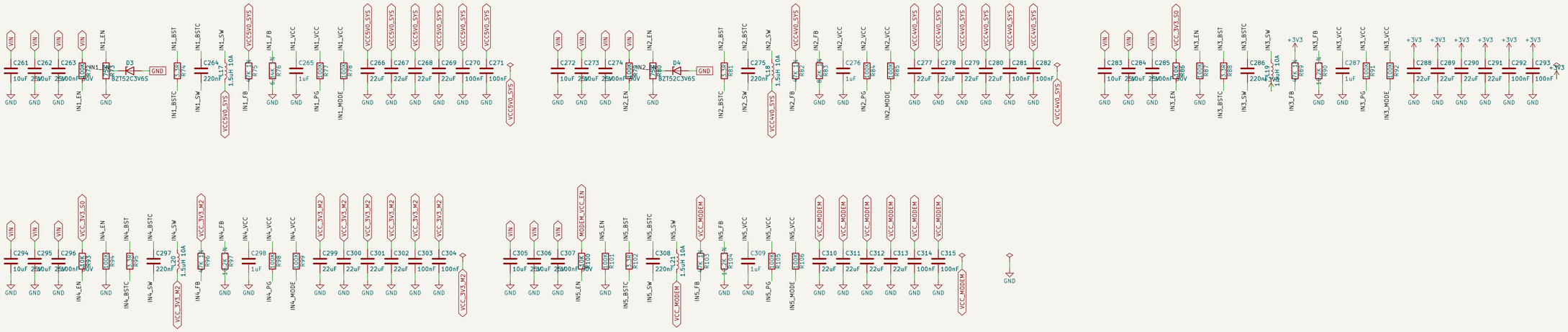












HookProbe
Sheet: /Input converters: 5 V, 4 V and three 3.3 V rails/
File: power_input.kicad_sch

Title: Input converters: 5 V, 4 V and three 3.3 V rails

Size: A2	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 11/25

Copied from the Radxa Rock 5B v1.45 schematic, sheets 4 and 12. Net names follow the reference, minus 'LP4/4X_'.
tools/verify_ddr.py confirms every memory-chip signal ball and its net against the reference page.
LPDDR4X only: VDDQ is tied to the 0.6 V rail, and VDD1 is taken straight from VCC_1V8_S3 (the reference
has zero-ohm option links there). The LPDDR5-only pins (WCK, A6) are left unconnected, as in the reference.
This interface must be laid out by copying Rockchip's memory layout template, not routed freehand.

RK3588
U100: DDR channel 0

U17

LPDDR4X 200-ball

HookProbe
Sheet: /Memory channel 0: LPDDR4X/
File: ddr0.kicad_sch
Title: Memory channel 0: LPDDR4X
Size: A2 | Date: | Rev: A0
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 | Id: 12/25

Copied from the Radxa Rock 5B v1.45 schematic, sheets 4 and 12. Net names follow the reference, minus 'LP4/4X_'.
tools/verify_ddr.py confirms every memory-chip signal ball and its net against the reference page.
LPDDR4X only: VDDQ is tied to the 0.6 V rail, and VDD1 is taken straight from VCC_1V8_S3 (the reference
has zero-ohm option links there). The LPDDR5-only pins (WCK, A6) are left unconnected, as in the reference.
This interface must be laid out by copying Rockchip's memory layout template, not routed freehand.

RK3588
U100: DDR channel 1

U18

LPDDR4X 200-ball

HookProbe
Sheet: /Memory channel 1: LPDDR4X/
File: ddr1.kicad_sch
Title: Memory channel 1: LPDDR4X
Size: A2 | Date: | Rev: A0
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 | Id: 13/25

RK3588 units 12 (system) and 6 (GPIO bank 0), from the Radxa Rock 5B v1.45 schematic, sheets 11, 13, 15 and 19.
IO bank voltages as the reference: banks on VCCIO5 and VCCIO6 run at 3.3 V, PMUIO2 at 3.3 V, the rest at 1.8 V.
Boot select (SARADC_IN0) has only the 100K pull-up: boot order SPI flash, eMMC, SD, USB. Shorting J to ground
forces USB download mode. No SD card slot and no SPI flash on this board, so their pins are left open.
PMUIO and memory VDD1 use VCC_1V8_S3 directly (the reference has a zero-ohm link to VCC1V8_PMU_DDR_S3).

RK3588
U100: System

VCC_1V8_S3 — V26 EMMCIO_1V8
RESET_L — M31 NPOR_u
VCCA_1V8_S0 — N27 OSC_1V8
VDD_0V75_S0 — AD3 OTP_VDDOTP_0V75
AD4 OTP_VPP
VCCA_1V8_S0 — U18 PLL_AVDD1V8
VDD_0V75_S0 — V20 PLL_DVDD0V75
VCC_1V8_S3 — N28 PMUIO1_1V8
VCC_3V3_S3 — P28 PMUIO2
VCC_1V8_S3 — R27 PMUIO2_1V8
VDD_0V75_S3 — P27 PMU_0V75
VCCA_1V8_S0 — AH18 SARADC_AVDD_1V8
BOOT_SARADC_IN0 — AM16 SARADC_IN0_BOOT
RECOVERY_KEY — AL16 SARADC_IN1
AK16 SARADC_IN2
AN17 SARADC_IN3
AM17 SARADC_IN4
AK15 SARADC_IN5
AL17 SARADC_IN6
AK17 SARADC_IN7
AF18 TSADC_TEST_OUT_TS
GND — M29 TVSS_d
VCC_1V8_S0 — G20 VCCIO1_1V8
VCCIO_SD_S0 — Y7 VCCIO2
VCC_1V8_S0 — AA7 VCCIO2_1V8
VCC_1V8_S0 — Y26 VCCIO3_1V8
VCC_1V8_S0 — H21 VCCIO4
VCC_1V8_S0 — H20 VCCIO4_1V8
VCC_3V3_S0 — W26 VCCIO5
VCC_1V8_S0 — W25 VCCIO5_1V8
VCC_3V3_S0 — AC26 VCCIO6
VCC_1V8_S0 — AC25 VCCIO6_1V8
XIN24M — R34 XIN_24M
XOUT24M — T34 XOUT_24M

RK3588
U100: GPIO bank 0

K29 CLK32K_IN/CLK32K_OUT0/GPIO0_B2_u
WIFI_DISABLE_L — R30 I2S1_LRCK_RX_M1/PDM0_CLK1_M1/PWM2_M0/UART0_RX_M0/I2C4_SDA_M2/DP0_HPDIN_M1/PCIE30X1_0_WAKEN_M0/GPIO0_C4_d
MODEM_RESET_H — P29 I2S1_LRCK_TX_M1/PWM0_M0/I2C2_SCL_M0/CAN0_TX_M0/SPI0_CS1_M0/PCIE30X1_1_PERSTN_M0/GPIO0_B7_d
UART2_TX_DEBUG — T31 I2S1_MCLK_M1/JTAG_TCK_M2/I2C1_SCL_M0/UART2_TX_M0/PCIE30X1_1_CLKREQN_M0/GPIO0_B5_d
FAN_PWM — R29 I2S1_SCLK_RX_M1/PDM0_CLK0_M1/PWM1_M0/I2C2_SDA_M0/CAN0_RX_M0/SPI0_MOSI_M0/PCIE30X1_0_CLKREQN_M0/GPIO0_C0_d
UART2_RX_DEBUG — R29 I2S1_SCLK_TX_M1/JTAG_TMS_M2/I2C1_SDA_M0/UART2_RX_M0/PCIE30X1_1_WAKEN_M0/GPIO0_B6_d
BT_DISABLE_L — P30 I2S1_SDI0_M1/GPU_AVS/UART0_TX_M0/I2C4_SCL_M2/DP1_HPDIN_M1/PWM4_M0/PCIE30X1_0_PERSTN_M0/GPIO0_C5_u
MODEM_DISABLE_L — T29 I2S1_SDI1_M1/NPU_AVS/UART0_RTSN_/PWM5_M1/SPI0_CLK_M0/PCIE30X4_CLKREQN_M0/SATA_CP_POD/GPIO0_C6_u
I2C6_SDA_M0 — V31 I2S1_SDI2_M1/PDM0_SDI0_M1/I2C6_SDA_M0/UART1_RTSN_M2/PWM6_M0/SPI0_MISO_M0/PCIE30X4_WAKEN_M0/GPIO0_C7_d
I2C6_SCL_M0 — W31 I2S1_SDI3_M1/PDM0_SDI1_M1/I2C6_SCL_M0/UART1_CTSN_M2/PWM7_IR_M0/SPI3_MISO_M2/PCIE30X4_PERSTN_M0/GPIO0_D0_d
I2C0_SCL_M2 — W30 I2S1_SDO0_M1/CPU_BIG0_AVS/I2C0_SCL_M2/UART0_CTSN/UART1_TX_M2/HDMI_RX_SDA_M0/SPI0_CS0_M0/PCIE30X2_CLKREQN_M0/HDMI_TX0_CEC_M1/GPIO0_D1_u
I2C0_SDA_M2 — W29 I2S1_SDO1_M1/I2C0_SDA_M2/UART1_RX_M2/HDMI_RX_SCL_M0/SPI3_MOSI_M2/PCIE30X2_WAKEN_M0/HDMI_TX1_CEC_M1/GPIO0_D2_u
I2C1_SCL_M2 — V29 I2S1_SDO2_M1/PDM0_SDI2_M1/PWM3_IR_M0/I2C1_SCL_M2/CAN2_RX_M1/HDMI_TX0_SDA_M1/SPI3_CS0_M2/PCIE30X2_PERSTN_M0/SATA_CPDET/GPIO0_D4_u
I2C1_SDA_M2 — V28 I2S1_SDO3_M1/CPU_BIG1_AVS/I2C1_SDA_M2/CAN2_TX_M1/HDMI_TX0_SCL_M1/SPI3_CS1_M2/SATA_MP_SWITCH/GPIO0_D5_u
MODEM_PWR_EN — U33 LITCPU_AVS/SPI3_CLK_M2/GPIO0_D3_u
PMIC_INT_L — M30 PMIC_INT_L/GPIO0_A7_u
GPIO0_A2_PIN — R32 PMIC_SLEEP1/GPIO0_A2_d
R31 PMIC_SLEEP2/GPIO0_A3_d
CPU_BIGO_VSEL — U32 PMIC_SLEEP3/GPIO0_C1_d
NPU_VSEL — T32 PMIC_SLEEP4/GPIO0_C2_d
GPIO0_C2_PIN — T32
GPIO0_C3_PIN — T30 PMIC_SLEEP5/GPIO0_C3_d
CPU_BIG1_VSEL — W28 PMIC_SLEEP6/PDM0_SDI3_M1/GPIO0_D6_d
P33 REFCLK_OUT/GPIO0_A0_d
P31 SDMMC_DET/GPIO0_A4_u
GPIO0_A5_PIN — N31 SPI2_CLK_M2/SDMMC_PWREN/PMU_DEBUG/GPIO0_A5_d
PMIC_SPI_CS — K30 SPI2_CS0_M2/I2C1_SDA_M1/PWM5_M0/UART0_TX_M1/GPIO0_B1_z
RTC_INT_L — L30 SPI2_CS1_M2/I2C1_SCL_M1/UART0_RX_M1/GPIO0_B0_z
PMIC_SPI_MOSI — L29 SPI2_MISO_M2/I2C0_SCL_M0/GPIO0_B3_z
GPIO0_A6_PIN — N30 SPI2_MOSI_M2/I2C0_SDA_M0/GPIO0_A6_z
TSADC_SHUT_H — P32 TSADC_SHUT_ORG/TSADC_SHUT/GPIO0_A1_z

Y3
24MHz
XTAL_X1 — 1
3 — XIN24M
2 — GND

BOOT_SARADC_IN0 — J9 1
GND — 2
USB download

XOUT24M — C429 100nF — GND
XOUT24M — 5.6K R435 — XIN24M
XTAL_X1 — C456 18pF — GND
XIN24M — C457 18pF — GND
RESET_L — C458 100nF — GND
RESET_L — R431 — TSADC_SHUT_H
VCCA_1V8_S0 — C459 100nF — BOOT_SARADC_IN0
VCCA_1V8_S0 — C460 100nF — RECOVERY_KEY
RECOVERY_KEY — C461 1nF — GND
GPIO0_A2_PIN — C462 100nF — PMIC_PWR_CTRL1
GPIO0_C2_PIN — C463 100nF — PMIC_PWR_CTRL3
PMIC_PWR_CTRL3 — C464 100nF — GND
GPIO0_C3_PIN — C465 100nF — PMIC_PWR_CTRL2
PMIC_PWR_CTRL2 — C466 100nF — GND
GPIO0_A5_PIN — C467 100nF — PMIC_SPI_CLK
GPIO0_A6_PIN — C468 100nF — PMIC_SPI_MOSI
VCC_3V3_S3 — C469 100nF — I2C0_SCL_M2
VCC_3V3_S3 — C470 100nF — I2C0_SDA_M2
VCC_3V3_S3 — C471 100nF — I2C1_SCL_M2
VCC_3V3_S3 — C472 100nF — I2C1_SDA_M2
VCC_3V3_S3 — C473 100nF — I2C6_SCL_M0
VCC_3V3_S3 — C474 100nF — I2C6_SDA_M0

VCC_1V8_S3 — C433 1uF — GND
VCCA_1V8_S0 — C434 100nF — GND
VCCA_1V8_S0 — C435 1uF — GND
VDD_0V75_S0 — C436 100nF — GND
VCCA_1V8_S0 — C437 100nF — GND
VDD_0V75_S0 — C438 100nF — GND
VDD_0V75_S0 — C439 1uF — GND
VCC_1V8_S3 — C440 100nF — GND
VCC_3V3_S3 — C441 100nF — GND
VCC_1V8_S3 — C442 100nF — GND
VDD_0V75_S3 — C443 100nF — GND
VDD_0V75_S3 — C444 1uF — GND
VCCA_1V8_S0 — C445 100nF — GND
VCC_1V8_S0 — C446 100nF — GND
VCCIO_SD_S0 — C447 100nF — GND
VCC_1V8_S0 — C448 100nF — GND
VCC_1V8_S0 — C449 100nF — GND
VCC_1V8_S0 — C450 100nF — GND
VCC_1V8_S0 — C451 100nF — GND
VCC_3V3_S0 — C452 100nF — GND
VCC_1V8_S0 — C453 100nF — GND
VCC_3V3_S0 — C454 100nF — GND
VCC_1V8_S0 — C455 100nF — GND

HookProbe
Sheet: /Processor: clock, reset, IO supplies, GPIO bank 0/
File: soc_system.kicad_sch

Title: Processor: clock, reset, IO supplies, GPIO bank 0

Size: A2	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 14/25

Bank 1 (1.8 V) is unused on this board, Bank 2 carries the eMMC interface, as the reference;
its other pins (1.8 V) are unused. The eMMC chip itself is not drawn yet.

RK3588
U100: GPIO bank 1

B26	HDMI_TX0_HPD_M0/SPI2_MOSI_M0/GPIO1_A5_d
C24	HDMI_TX1_HPD_M0/SPI2_CLK_M0/GPIO1_A6_d
B25	HDMI_TX1_SCL_M2/SPI2_MISO_M0/GPIO1_A4_d
A27	HDMI_TX1_SDA_M2/I2C4_SCL_M3/UART6_CTSN_M1/PWM1_M2/SPI4_CS0_M2/GPIO1_A3_d
D27	I2C3_SCL_M0/UART3_TX_M0/SPI4_MOSI_M0/GPIO1_C1_z
G29	I2C3_SDA_M0/UART3_RX_M0/SPI4_MISO_M0/GPIO1_C0_z
D29	I2S0_LRCK_RX/PDM0_CLK0_M0/I2C4_SDA_M4/PWM15_IR_M2/GPIO1_C6_d
D30	I2S0_LRCK_TX/I2C2_SCL_M3/UART4_RTSN/GPIO1_C5_d
F30	I2S0_MCLK/I2C6_SDA_M1/UART3_RTSN/PWM3_IR_M2/SPI4_CLK_M0/GPIO1_C2__d
E30	I2S0_SCLK_RX/PDM0_CLK1_M0/I2C2_SDA_M3/PWM11_IR_M2/SPI4_CS1_M0/GPIO1_C4_d
E31	I2S0_SCLK_TX/I2C6_SCL_M1/UART3_CTSN/PWM7_IR_M2/SPI4_CS0_M0/GPIO1_C3_d
D28	I2S0_SDI0/GPIO1_D4_d
E28	I2S0_SDI1/PDM0_SDI3_M0/I2C1_SDA_M4/UART4_RX_M0/PWM1_M1/SPI1_CS0_M2/GPIO1_D3_d
E29	I2S0_SDO0/I2C4_SCL_M4/UART4_CTSN/GPIO1_C7_d
F26	I2S0_SDO1/I2C7_SCL_M0/UART6_TX_M2/SPI1_MISO_M2/GPIO1_D0_d
F27	I2S0_SDO2/I2S0_SDI3/PDM0_SDI1_M0/I2C7_SDA_M0/UART6_RX_M2/SPI1_MOSI_M2/GPIO1_D1_d
F28	I2S0_SDO3/I2S0_SDI2/PDM0_SDI2_M0/I2C1_SCL_M4/UART4_TX_M0/PWM0_M1/SPI1_CLK_M2/GPIO1_D2_d
E26	MIPI_CAMERA1_CLK_M0/SPDIF0_TX_M0/PCIE30X2_WAKEN_M3/HDMI_RX_HPDIN_M2/I2C5_SCL_M3/UART1_TX_M1/GPIO1_B6_d
E27	MIPI_CAMERA2_CLK_M0/SPDIF1_TX_M0/PCIE30X2_PERSTN_M3/HDMI_RX_CEC_M2/SATA2_ACT_LED_M1/I2C5_SDA_M3/UART1_RX_M1/PWM13_M2/GPIO1_B7_u
F24	MIPI_CAMERA3_CLK_M0/HDMI_RX_SCL_M2/I2C8_SCL_M2/UART1_RTSN_M1/PWM14_M2/GPIO1_D6_u
F25	MIPI_CAMERA4_CLK_M0/PCIE30X2_CLKREQN_M3/HDMI_RX_SDA_M2/I2C8_SDA_M2/UART1_CTSN_M1/PWM15_IR_M3/GPIO1_D7_u
E25	PCIE30X1_0_CLKREQN_M2/UART7_TX_M2/SPI0_CS1_M2/GPIO1_B5_u
A24	PCIE30X1_1_CLKREQN_M2/DP0_HPDIN_M2/I2C2_SDA_M4/UART6_RX_M1/SPI4_MISO_M2/GPIO1_A0_d
A25	PCIE30X1_1_WAKEN_M2/DP1_HPDIN_M2/SATA1_ACT_LED_M1/I2C2_SCL_M4/UART6_TX_M1/SPI4_MOSI_M2/GPIO1_A1_d
G26	PDM0_SDI0_M0/SPI1_CS1_M2/GPIO1_D5_d
E24	PDM1_CLK0_M1/PCIE30X1_0_PERSTN_M2/UART7_RX_M2/SPI0_CS0_M2/GPIO1_B4_u
D27	PDM1_CLK1_M1/PCIE30X1_0_WAKEN_M2/SATA0_ACT_LED_M1/UART4_TX_M2/SPI0_CLK_M2/GPIO1_B3_d
C25	PDM1_SDI0_M1/PCIE30X1_1_PERSTN_M2/PWM3_IR_M3/SPI2_CS0_M0/GPIO1_A7_u
C27	PDM1_SDI1_M1/PCIE30X4_CLKREQN_M3/SPI2_CS1_M0/GPIO1_B0_u
D25	PDM1_SDI2_M1/PCIE30X4_WAKEN_M3/SPI0_MISO_M2/GPIO1_B1_d
D26	PDM1_SDI3_M1/PCIE30X4_PERSTN_M3/UART4_RX_M2/SPI0_MOSI_M2/GPIO1_B2_d
A26	VOP_POST_EMPTY/I2C4_SDA_M3/UART6_RTSN_M1/PWM0_M2/SPI4_CLK_M2/GPIO1_A2_d

RK3588
U100: GPIO bank 2

AE30	CLK32K_OUT1/GPIO2_C5_d
V34	EMMC_CLKOUT/GPIO2_A1_d
W34	EMMC_CMD/FSPI_CLK_M0/GPIO2_A0_u
Y33	EMMC_D0/FSPI_D0_M0/GPIO2_D0_u
W33	EMMC_D1/FSPI_D1_M0/GPIO2_D1_u
V32	EMMC_D2/FSPI_D2_M0/GPIO2_D2_u
AA33	EMMC_D3/FSPI_D3_M0/GPIO2_D3_u
Y32	EMMC_D4/I2C1_SCL_M3/UART5_RX_M2/GPIO2_D4_u
AA32	EMMC_D5/I2C1_SDA_M3/UART5_TX_M2/GPIO2_D5_u
W32	EMMC_D6/FSPI_CS0N_M0/GPIO2_D6_u
V33	EMMC_D7/FSPI_CS1N_M0/GPIO2_D7_u
Y34	EMMC_DATA_STROBE/I2C2_SDA_M2/UART5_CTSN_M1/GPIO2_A2_d
AA34	EMMC_RSTN/I2C2_SCL_M2/UART5_RTSN_M1/GPIO2_A3_d
AD30	ETH0_REFCLKO_25M/I2S2_SDI_M0/I2C6_SCL_M2/SPI1_CS0_M0/GPIO2_C3_d
AC30	GMAC0_PPSCLK/TEST_CLKOUT_M1/HDMI_TX1_CEC_M0/UART9_RX_M0/SPI1_CS1_M0/GPIO2_C4_d
AB30	GMAC0_PPSTRING/FSPI_CS1N_M1/HDMI_TX1_SCL_M0/I2C4_SCL_M1/UART7_TX_M0/GPIO2_B5_u
AB31	GMAC0_PTP_REFCLK/FSPI_CS0N_M1/HDMI_TX1_SDA_M0/I2C4_SDA_M1/UART7_RX_M0/GPIO2_B4_u
AE32	GMAC0_RXCLK/SDIO_D2_M0/FSPI_D2_M1/I2C8_SCL_M1/UART6_RTSN_M0/GPIO2_B0_u
AD32	GMAC0_RXD0/I2S2_SCLK_RX_M0/I2C2_SCL_M1/UART1_CTSN_M0/SPI1_MISO_M0/GPIO2_C1_d
AD31	GMAC0_RXD1/I2S2_LRCK_RX_M0/I2C6_SDA_M2/UART9_TX_M0/SPI1_MOSI_M0/GPIO2_C2_d
AC32	GMAC0_RXD2/SDIO_D0_M0/FSPI_D0_M1/UART6_RX_M0/GPIO2_A6_u
AC31	GMAC0_RXD3/SDIO_D1_M0/FSPI_D1_M1/UART6_TX_M0/GPIO2_A7_u
AE33	GMAC0_TXCLK/SDIO_CLK_M0/FSPI_CLK_M1/I2C3_SDA_M3/GPIO2_B3_d
AD33	GMAC0_TXD0/I2S2_MCLK_M0/I2C5_SCL_M4/UART1_RX_M0/GPIO2_B6_d
AD34	GMAC0_TXD1/I2S2_SCLK_TX_M0/I2C5_SDA_M4/UART1_TX_M0/GPIO2_B7_d
AC33	GMAC0_TXD2/SDIO_D3_M0/FSPI_D3_M1/I2C8_SDA_M1/UART6_CTSN_M0/GPIO2_B1_u
AC34	GMAC0_TXD3/SDIO_CMD_M0/I2C3_SCL_M3/GPIO2_B2_u
AE34	GMAC0_TXEN/I2S2_LRCK_TX_M0/I2C2_SDA_M1/UART1_RTSN_M0/SPI1_CLK_M0/GPIO2_C0_d

HookProbe	
Sheet: /Processor: GPIO banks 1 and 2/	
File: soc_gpio12.kicad_sch	
Title: Processor: GPIO banks 1 and 2	
Size: A2	Date:
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8	Rev: A0 Id: 15/25

Banks 3 and 4 run at 3.3 V (pins GPIO4_C2–C6 at 1.8 V, GPIO4_D0–D5 on the SD supply: unused).
PCIe reset and clock-request lines sit on each controller's own M1 pins. Assignments are in GPIO_USE
in build.py; those marked 'this board' are new and not from the reference.

RK3588
U100: GPIO bank 3

	AH26	CIF_D8/FSPI_CSON_M2/PCIE3OX4_CLKREQN_M2/HDMI_TX1_CEC_M2/CAN2_RX_M0/UART5_TX_M1/SPI3_CS0_M3/GPIO3_C4_u
	AH25	CIF_D9/FSPI_CS1N_M2/PCIE3OX4_WAKEN_M2/HDMI_TX1_SDA_M1/CAN2_TX_M0/UART5_RX_M1/SPI3_CS1_M3/GPIO3_C5_u
	AG26	CIF_D10/PCIE3OX4_PERSTN_M2/HDMI_TX1_SCL_M1/SPI3_MISO_M3/GPIO3_C6_u
PCIE_WIFI_CLKREQ_L	AJ24	CIF_D11/PCIE2OX1_2_CLKREQN_M0/HDMI_TX0_SCL_M2/I2C5_SCL_M0/SPI3_MOSI_M3/GPIO3_C7_u
	AH24	CIF_D12/PCIE2OX1_2_WAKEN_M0/HDMI_TX0_SDA_M2/I2C5_SDA_M0/UART4_RX_M1/PWM8_M2/SPI3_CLK_M3/GPIO3_D0_u
	AG23	CIF_D13/PCIE2OX1_2_PERSTN_M0/HDMI_RX_CEC_M1/UART4_TX_M1/PWM9_M2/SPI0_MISO_M3/GPIO3_D1_d
	AG25	CIF_D14/PCIE3OX2_CLKREQN_M2/HDMI_RX_SCL_M1/I2C7_SCL_M2/UART9_RTSN_M2/SPI0_MOSI_M3/GPIO3_D2_d
	AG24	CIF_D15/PCIE3OX2_WAKEN_M2/HDMI_RX_SDA_M1/I2C7_SDA_M2/UART9_CTSN_M2/PWM10_M2/SPI0_CLK_M3/GPIO3_D3_d
THREAD_RESET_L	AH27	ETH1_REFCLKO_25M/MIPI_CAMERA1_CLK_M1/I2C4_SCL_M0/GPIO3_A6_d
UART3_RX_THREAD	AE29	GMAC1_MCLKINOUT/I2S2_LRCK_TX_M1/CAN1_TX_M0/UART3_RX_M1/PWM13_M0/GPIO3_B6_d
SPI1_CS0_TPM	Y31	GMAC1_MDC/MIPI_TE0/I2C8_SCL_M4/UART7_RTSN_M1/PWM14_M0/SPI1_CS0_M1/GPIO3_C2_d
TPM_INT_L	Y30	GMAC1_MDIO/MIPI_TE1/I2C8_SDA_M4/UART7_CTSN_M1/PWM15_IR_M0/SPI1_CS1_M1/GPIO3_C3_d
SPI1_CLK_TPM	Y27	GMAC1_PPSCLK/PCIE3OX2_BUTTON_RSTN/UART7_RX_M1/SPI1_CLK_M1/GPIO3_C1_d
SPI1_MISO_TPM	Y29	GMAC1_PPSTRIG/I2C3_SDA_M1/UART7_TX_M1/SPI1_MISO_M1/GPIO3_C0_d
SPI1_MOSI_TPM	AA28	GMAC1_PTP_REF_CLK/HDMI_TX1_HPD_M1/I2C3_SCL_M1/SPI1_MOSI_M1/GPIO3_B7_d
	AH30	GMAC1_RXCLK/SDIO_CLK_M1/MIPI_CAMERA0_CLK_M1/FSPI_CLK_M2/I2C4_SDA_M0/UART8_CTSN_M1/GPIO3_A5_d
THREAD_BOOT	AG29	GMAC1_RXD0/MIPI_CAMERA2_CLK_M1/PWM8_M0/GPIO3_A7_u
PCIE_WIFI_PERST_L	AG28	GMAC1_RXD1/I2S2_SCLK_RX_M1/MIPI_CAMERA3_CLK_M1/PWM9_M0/GPIO3_B0_u
	AD27	GMAC1_RXD2/SDIO_D2_M1/I2S3_LRCK/AUDDSM_LP/FSPI_D2_M2/UART8_TX_M1/SPI4_CLK_M1/GPIO3_A2_u
	AE27	GMAC1_RXD3/SDIO_D3_M1/I2S3_SDO/AUDDSM_RN/FSPI_D3_M2/UART8_RX_M1/SPI4_CS0_M1/GPIO3_A3_u
MODEM_VCC_EN	AH29	GMAC1_RXDV_CRS/I2S2_LRCK_RX_M1/MIPI_CAMERA4_CLK_M1/UART2_TX_M2/PWM2_M1/GPIO3_B1_d
	AD28	GMAC1_TXCLK/SDIO_CMD_M1/I2S3_SDI/AUDDSM_RP/UART8_RTSN_M1/SPI4_CS1_M1/GPIO3_A4_d
	AC28	GMAC1_TXD0/I2S2_SDO_M1/UART2_RTSN/GPIO3_B3_u
PD1_INT_L	AC29	GMAC1_TXD1/I2S2_MCLK_M1/UART2_CTSN/GPIO3_B4_u
	AA29	GMAC1_TXD2/SDIO_D0_M1/I2S3_MCLK/FSPI_D0_M2/I2C6_SDA_M4/PWM10_M0/SPI4_MISO_M1/GPIO3_A0_u
	AA30	GMAC1_TXD3/SDIO_D1_M1/I2S3_SCLK/AUDDSM_LN/FSPI_D1_M2/I2C6_SCL_M4/PWM11_IR_M0/SPI4_MOSI_M1/GPIO3_A1_u
UART3_TX_THREAD	AD29	GMAC1_TXEN/I2S2_SCLK_TX_M1/CAN1_RX_M0/UART3_TX_M1/PWM12_M0/GPIO3_B5_u
PD2_INT_L	AE28	GMAC1_TXER/I2S2_SDI_M1/UART2_RX_M2/PWM3_IR_M1/GPIO3_B2_d
	AA27	HDMI_TX0_HPD_M1/PCIE3OX2_PERSTN_M2/HDMI_RX_HPDIN_M1/MCU_JTAG_TCK_M1/UART9_RX_M2/SPI0_CS0_M3/GPIO3_D4_d
	AB28	PCIE3OX4_BUTTON_RSTN/DP1_HPDIN_M0/MCU_JTAG_TMS_M1/UART9_TX_M2/PWM11_IR_M3/SPI0_CS1_M3/GPIO3_D5_d

RK3588
U100: GPIO bank 4

PCIE_WAKE_L	AJ26	BT1120_D11/PCIE3OX4_WAKEN_M1/HDMI_RX_CEC_M0/SATA1_ACT_LED_M0/UART9_RX_M1/PWM12_M1/SPI3_MISO_M1/GPIO4_B5_d
PCIE_NVME_PERST_L	AJ27	BT1120_D12/PCIE3OX4_PERSTN_M1/HDMI_RX_HPDIN_M0/SATA0_ACT_LED_M0/I2C5_SCL_M1/PWM13_M1/SPI3_MOSI_M1/GPIO4_B6_d
USBC2_SBU1_DC	AJ28	BT1120_D13/PCIE2OX1_2_CLKREQN_M1/HDMI_TX0_SCL_M0/I2C5_SDA_M1/SPI3_CLK_M1/GPIO4_B7_u
USBC2_SBU2_DC	AJ25	BT1120_D14/PCIE2OX1_2_WAKEN_M1/HDMI_TX0_SDA_M0/I2C8_SCL_M3/SPI3_CS0_M1/GPIO4_C0_u
PCIE_ACCEL_CLKREQ_L	AK24	BT1120_D15/SPDIF1_TX_M2/PCIE2OX1_2_PERSTN_M1/HDMI_TX0_CEC_M0/I2C8_SDA_M3/PWM6_M1/SPI3_CS1_M1/GPIO4_C1_d
PCIE_ACCEL_PERST_L	AK26	CIF_CLKIN/BT1120_CLKOUT/I2S1_SDI3_M0/PCIE30X2_PERSTN_M1/I2C6_SDA_M3/UART8_TX_M0/SPI2_CS1_M1/GPIO4_B0_d
PCIE_NVME_CLKREQ_L	AL26	CIF_CLKOUT/BT1120_D10/I2S1_SDO3_M0/PCIE30X4_CLKREQN_M1/DP0_HPDIN_M0/SPDIF0_TX_M1/UART9_TX_M1/PWM11_IR_M1/GPIO4_B4_u
PCIE_ETH2_CLKREQ_L	AK30	CIF_D0/BT1120_D0/I2S1_MCLK_M0/PCIE30X1_1_CLKREQN_M1/UART9_RTSN_M1/SPI0_MISO_M1/GPIO4_A0_d
	AL30	CIF_D1/BT1120_D1/I2S1_SCLK_TX_M0/PCIE30X1_1_WAKEN_M1/UART9_CTSN_M1/SPI0_MOSI_M1/GPIO4_A1_d
PCIE_ETH2_PERST_L	AM29	CIF_D2/BT1120_D2/I2S1_LRCK_TX_M0/PCIE30X1_1_PERSTN_M1/SPI0_CLK_M1/GPIO4_A2_d
PCIE_ETH1_CLKREQ_L	AL29	CIF_D3/BT1120_D3/I2S1_SCLK_RX_M0/PCIE30X1_0_CLKREQN_M1/UART0_TX_M2/GPIO4_A3_d
USBC2_VBUS_EN	AL28	CIF_D4/BT1120_D4/I2S1_LRCK_RX_M0/PCIE30X1_0_WAKEN_M1/I2C3_SCL_M2/UART0_RX_M2/SPI2_MISO_M1/GPIO4_A4_d
PCIE_ETH1_PERST_L	AK27	CIF_D5/BT1120_D5/I2S1_SDIO_M0/PCIE30X1_0_PERSTN_M1/I2C3_SDA_M2/UART3_TX_M2/SPI2_MOSI_M1/GPIO4_A5_d
USBC1_SBU1_DC	AL27	CIF_D6/BT1120_D6/I2S1_SDI1_M0/PCIE30X2_CLKREQN_M1/I2C5_SCL_M2/UART3_RX_M2/SPI2_CLK_M1/GPIO4_A6_d
USBC1_SBU2_DC	AM27	CIF_D7/BT1120_D7/I2S1_SDI2_M0/PCIE30X2_WAKEN_M1/I2C5_SDA_M2/SPI2_CS0_M1/GPIO4_A7_d
I2C7_SCL_M3	AK29	CIF_HREF/BT1120_DB/I2S1_SDO1_M0/PCIE30X1_1_BUTTON_RSTN/I2C7_SCL_M3/UART8_RTSN_M0/PWM14_M1/SPI0_CS0_M1/CAN1_RX_M1/GPIO4_B2_u
I2C7_SDA_M3	AM28	CIF_VSYNC/BT1120_D9/I2S1_SDO2_M0/PCIE20X1_2_BUTTON_RSTN/I2C7_SDA_M3/UART8_CTSN_M0/PWM15_IR_M1/CAN1_TX_M1/GPIO4_B3_u
	AF34	GMAC0_MCLKINOUT/I2S2_SDO_M0/I2C7_SCL_M1/PWM4_M1/SPI3_CS1_M0/GPIO4_C3_d
	AB34	GMAC0_MDC/I2C7_SDA_M1/UART9_RTSN_M0/PWM5_M2/SPI3_MISO_M0/GPIO4_C4_d
	AB33	GMAC0_MDIO/I2C0_SCL_M1/UART9_CTSN_M0/PWM6_M2/SPI3_MOSI_M0/GPIO4_C5_d
	AE31	GMAC0_RXDV_CRS/UART7_RTSN_M0/PWM2_M2/SPI3_CS0_M0/GPIO4_C2_d
	AF33	GMAC0_TXER/I2C0_SDA_M1/UART7_CTSN_M0/PWM7_IR_M3/SPI3_CLK_M0/GPIO4_C6_d
USB_HOST_PWREN_H	AL24	MIPI_CAMERA0_CLK_M0/SPDIF1_TX_M1/I2S1_SDO0_M0/PCIE30X1_0_BUTTON_RSTN/SATA2_ACT_LED_M0/I2C6_SCL_M3/UART8_RX_M0/SPI0_CS1_M1/GPIO4_B1_u
	AE1	SDMMC_CLK/PDM1_CLK0_M0/TEST_CLKOUT_M0/MCU_JTAG_TMS_M0/CAN0_RX_M1/UART5_TX_M0/GPIO4_D5_d
	AE2	SDMMC_CMD/PDM1_CLK1_M0/MCU_JTAG_TCK_M0/CAN0_TX_M1/UART5_RX_M0/PWM7_IR_M1/GPIO4_D4_u
	AD2	SDMMC_D0/PDM1_SDI3_M0/JTAG_TCK_M1/I2C3_SCL_M4/UART2_TX_M1/PWM8_M1/GPIO4_D0_u
	AD1	SDMMC_D1/PDM1_SDI2_M0/JTAG_TMS_M1/I2C3_SDA_M4/UART2_RX_M1/PWM9_M1/GPIO4_D1_u
	AF2	SDMMC_D2/PDM1_SDI1_M0/JTAG_TCK_M0/I2C8_SCL_M0/UART5_CTSN_M0/GPIO4_D2_u
	AF1	SDMMC_D3/PDM1_SDI0_M0/JTAG_TMS_M0/I2C8_SDA_M0/UART5_RTSN_M0/PWM10_M1/GPIO4_D3_u

VCC_3V3_SD
R249 2.2K
I2C7_SDA_M3
VCC_3V3_SD
R247 2.2K
I2C7_SCL_M3

HookProbe
Sheet: /Processor: GPIO banks 3 and 4/
File: soc_gpio34.kicad_sch
Title: Processor: GPIO banks 3 and 4
Size: A2 | Date: | Rev: A0
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8 | Id: 16/25

Each buffer circuit is copied from the Radxa Rock 5B v1.45 schematic, sheet 7 (Au5426, same pinout as TI LMK00304).
Straps as the reference: input 0 selected, HCSL outputs, reference output off. Outputs have 33R in series
and 49.9R to ground on each leg. Differences from the reference, to check against the buffer datasheet:
  – two buffers share the one oscillator (the reference has a single buffer);
  – each regulator is always enabled (the reference switches it from a GPIO);
  – the zero-ohm links between the regulator and the four supply pins groups are left out;
  – two outputs of buffer B are unused and left open.
The WI-FI slot does not use this clock: its reference clock comes from the processor's combo PHY.

U19 RT9193-33GB

U21 RT9193-33GB

U20

U22

Au5426 / LMK00304

Au5426 / LMK00304

Y4 OSC_VDD 100MHz 3.3V 3225

FB4 120R@100MHz 0.6A

HookProbe		
Sheet: /PCIe reference clock: oscillator and two buffers/		
File: pcie_clock.kicad_sch		
Title: PCIe reference clock: oscillator and two buffers		
Size: A2	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 17/25

RK3588 unit 5: HDMI/eDP transmitters, HDMI receiver, MIPI CSI/DSI. None is used on this board
(setup video goes out over USB-C DisplayPort from the Type-C PHYs).
Supply pins are powered from the same rails as on the Radxa Rock 5B v1.45 schematic (sheets 16, 17), with one
100 nF each; all signal, REXT and VREG pins are left open. Two rails were not readable from the reference and are
assigned by voltage (HDMI/eDP AVDD_0V75, HDMI_RX_DVDD). Confirm this unused-block treatment against
Rockchip's hardware design guide before layout.

RK3588
U100: Display and camera

VDDA_0V75_S0	AB9	HDMI/eDP_TX0_AVDD_0V75
	AM2	HDMI/eDP_TX0_REXT
VDDA_0V75_S0	AA9	HDMI/eDP_TX0_VDD_0V75
VCC_1V8_S0	AC6	HDMI/eDP_TX0_VDD_CMN_1V8
VCC_1V8_S0	AC7	HDMI/eDP_TX0_VDD_IO_1V8
VDDA_0V75_S0	AC9	HDMI/eDP_TX1_AVDD_0V75
	AN1	HDMI/eDP_TX1_REXT
VDDA_0V75_S0	AD9	HDMI/eDP_TX1_VDD_0V75
VCC_1V8_S0	AD6	HDMI/eDP_TX1_VDD_CMN_1V8
VCC_1V8_S0	AD7	HDMI/eDP_TX1_VDD_IO_1V8
VDDA_0V75_S0	AE8	HDMI_RX_AVDD0V75
	AF5	HDMI_RX_CLKN
	AF6	HDMI_RX_CLKP
	AG4	HDMI_RX_D0N
	AG5	HDMI_RX_D0P
	AH5	HDMI_RX_D1N
	AH6	HDMI_RX_D1P
	AJ4	HDMI_RX_D2N
	AJ5	HDMI_RX_D2P
VCCA_3V3_S0	AE9	HDMI_RX_DVDD3V3
	AF3	HDMI_RX_REXT
VCCA_3V3_S0	AE4	HDMI_RX_VPH3V3
	AJ1	HDMI_TX0_D0N/eDP_TX0_D0N
	AJ2	HDMI_TX0_D0P/eDP_TX0_D0P
	AK2	HDMI_TX0_D1N/eDP_TX0_D1N
	AK3	HDMI_TX0_D1P/eDP_TX0_D1P
	AL1	HDMI_TX0_D2N/eDP_TX0_D2N
	AL2	HDMI_TX0_D2P/eDP_TX0_D2P
	AH2	HDMI_TX0_D3N/eDP_TX0_D3N
	AH3	HDMI_TX0_D3P/eDP_TX0_D3P
	AG1	HDMI_TX0_SBDN/eDP_TX0_AUXN
	AG2	HDMI_TX0_SBDP/eDP_TX0_AUXP
	AP4	HDMI_TX1_D0N/eDP_TX1_D0N
	AN4	HDMI_TX1_D0P/eDP_TX1_D0P
	AN5	HDMI_TX1_D1N/eDP_TX1_D1N
	AM5	HDMI_TX1_D1P/eDP_TX1_D1P
	AP6	HDMI_TX1_D2N/eDP_TX1_D2N
	AN6	HDMI_TX1_D2P/eDP_TX1_D2P
	AN3	HDMI_TX1_D3N/eDP_TX1_D3N
	AM3	HDMI_TX1_D3P/eDP_TX1_D3P
	AP2	HDMI_TX1_SBDN/eDP_TX1_AUXN
	AN2	HDMI_TX1_SBDP/eDP_TX1_AUXP
VDD_0V75_S0	AB26	MIPI_CSI0_AVCC0V75
VCC_1V8_S0	AB26	MIPI_CSI0_AVCC1V8
	AJ34	MIPI_CSI0_CLK0N
	AJ33	MIPI_CSI0_CLK0P
	AM34	MIPI_CSI0_CLK1N
	AM33	MIPI_CSI0_CLK1P
	AG34	MIPI_CSI0_D0N
	AG33	MIPI_CSI0_D0P
	AH34	MIPI_CSI0_D1N
	AH33	MIPI_CSI0_D1P
	AK34	MIPI_CSI0_D2N
	AK33	MIPI_CSI0_D2P
	AL34	MIPI_CSI0_D3N
	AL33	MIPI_CSI0_D3P
VDD_0V75_S0	AA25	MIPI_CSI1_AVCC0V75
VCC_1V8_S0	AA26	MIPI_CSI1_AVCC1V8
	AJ32	MIPI_CSI1_CLK0N
	AJ31	MIPI_CSI1_CLK0P
	AM32	MIPI_CSI1_CLK1N
	AM31	MIPI_CSI1_CLK1P
	AG32	MIPI_CSI1_D0N
	AG31	MIPI_CSI1_D0P
	AH32	MIPI_CSI1_D1N
	AH31	MIPI_CSI1_D1P
	AK32	MIPI_CSI1_D2N
	AK31	MIPI_CSI1_D2P
	AL32	MIPI_CSI1_D3N
	AL31	MIPI_CSI1_D3P
VDD_0V75_S0	AG20	MIPI_D/C_PHY0_VDD
VDDA_1V2_S0	AH20	MIPI_D/C_PHY0_VDD_1V2
VCC_1V8_S0	AJ20	MIPI_D/C_PHY0_VDD_1V8
	AF20	MIPI_D/C_PHY0_VREG
VDD_0V75_S0	AG19	MIPI_D/C_PHY1_VDD
VDDA_1V2_S0	AH19	MIPI_D/C_PHY1_VDD_1V2
VCC_1V8_S0	AJ19	MIPI_D/C_PHY1_VDD_1V8
	AF19	MIPI_D/C_PHY1_VREG
	AP31	MIPI_DPHY0_RX_CLKN/MIPI_CPHY0_RX_TRIO1_B
	AN32	MIPI_DPHY0_RX_CLKP/MIPI_CPHY0_RX_TRIO1_C
	AP29	MIPI_DPHY0_RX_D0N/MIPI_CPHY0_RX_TRIO0_A
	AN29	MIPI_DPHY0_RX_D0P/MIPI_CPHY0_RX_TRIO0_B
	AP30	MIPI_DPHY0_RX_D1N/MIPI_CPHY0_RX_TRIO0_C
	AN30	MIPI_DPHY0_RX_D1P/MIPI_CPHY0_RX_TRIO1_A
	AP32	MIPI_DPHY0_RX_D2N/MIPI_CPHY0_RX_TRIO2_A
	AN33	MIPI_DPHY0_RX_D2P/MIPI_CPHY0_RX_TRIO2_B
	AP33	MIPI_DPHY0_RX_D3N/MIPI_CPHY0_RX_TRIO2_C
	AN34	MIPI_DPHY0_RX_D3P/NO_USE
	AP26	MIPI_DPHY0_TX_CLKN/MIPI_CPHY0_TX_TRIO1_B
	AN26	MIPI_DPHY0_TX_CLKP/MIPI_CPHY0_TX_TRIO1_C
	AP24	MIPI_DPHY0_TX_D0N/MIPI_CPHY0_TX_TRIO0_A
	AN24	MIPI_DPHY0_TX_D0P/MIPI_CPHY0_TX_TRIO0_B
	AP25	MIPI_DPHY0_TX_D1N/MIPI_CPHY0_TX_TRIO0_C
	AN25	MIPI_DPHY0_TX_D1P/MIPI_CPHY0_TX_TRIO1_A
	AP27	MIPI_DPHY0_TX_D2N/MIPI_CPHY0_TX_TRIO2_A
	AN27	MIPI_DPHY0_TX_D2P/MIPI_CPHY0_TX_TRIO2_B
	AP28	MIPI_DPHY0_TX_D3N/MIPI_CPHY0_TX_TRIO2_C
	AN28	MIPI_DPHY0_TX_D3P/NO_USE
	AL20	MIPI_DPHY1_RX_CLKN/MIPI_CPHY1_RX_TRIO1_B
	AK20	MIPI_DPHY1_RX_CLKP/MIPI_CPHY1_RX_TRIO1_C
	AL18	MIPI_DPHY1_RX_D0N/MIPI_CPHY1_RX_TRIO0_A
	AK18	MIPI_DPHY1_RX_D0P/MIPI_CPHY1_RX_TRIO0_B
	AL19	MIPI_DPHY1_RX_D1N/MIPI_CPHY1_RX_TRIO0_C
	AK19	MIPI_DPHY1_RX_D1P/MIPI_CPHY1_RX_TRIO1_A
	AL21	MIPI_DPHY1_RX_D2N/MIPI_CPHY1_RX_TRIO2_A
	AK21	MIPI_DPHY1_RX_D2P/MIPI_CPHY1_RX_TRIO2_B
	AL22	MIPI_DPHY1_RX_D3N/MIPI_CPHY1_RX_TRIO2_C
	AK22	MIPI_DPHY1_RX_D3P/NO_USE
	AP20	MIPI_DPHY1_TX_CLKN/MIPI_CPHY1_TX_TRIO1_B
	AN20	MIPI_DPHY1_TX_CLKP/MIPI_CPHY1_TX_TRIO1_C
	AP18	MIPI_DPHY1_TX_D0N/MIPI_CPHY1_TX_TRIO0_A
	AN18	MIPI_DPHY1_TX_D0P/MIPI_CPHY1_TX_TRIO0_B
	AP19	MIPI_DPHY1_TX_D1N/MIPI_CPHY1_TX_TRIO0_C
	AN19	MIPI_DPHY1_TX_D1P/MIPI_CPHY1_TX_TRIO1_A
	AP21	MIPI_DPHY1_TX_D2N/MIPI_CPHY1_TX_TRIO2_A
	AN21	MIPI_DPHY1_TX_D2P/MIPI_CPHY1_TX_TRIO2_B
	AP22	MIPI_DPHY1_TX_D3N/MIPI_CPHY1_TX_TRIO2_C
	AN22	MIPI_DPHY1_TX_D3P/NO_USE

VDDA_0V75_S0	VDDA_0V75_S0	VCC_1V8_S0	VCC_1V8_S0	VDDA_0V75_S0	VDDA_0V75_S0	VCC_1V8_S0	VCC_1V8_S0	VDDA_0V75_S0	VCCA_3V3_S0	VCCA_3V3_S0	VDD_0V75_S0	VCC_1V8_S0	VDD_0V75_S0	VCC_1V8_S0	VDD_0V75_S0	VDDA_1V2_S0	VCC_1V8_S0	VDD_0V75_S0	VDDA_1V2_S0	VCC_1V8_S0							
C510	C511	C512	C513	C514	C515	C516	C517	C518	C519	C520	C521	C522	C523	C524	C525	C526	C527	C528	C529	C530							
100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF	100nF							
GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND	GND							

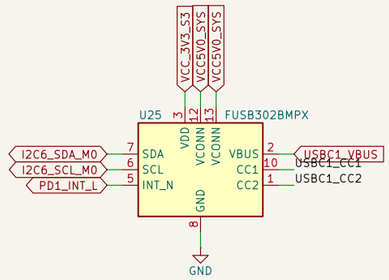
HookProbe

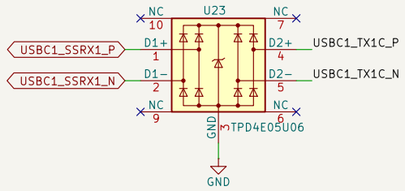
Sheet: /Processor: display and camera blocks (unused)/
File: soc_display.kicad_sch

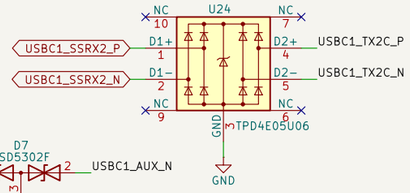
Title: Processor: display and camera blocks (unused)

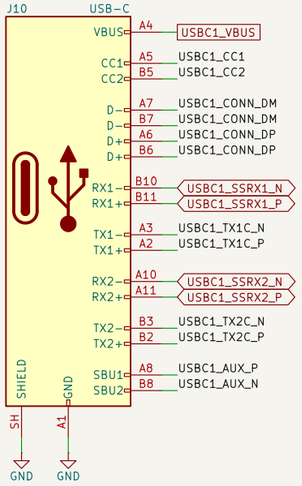
Size: A2	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 18/25

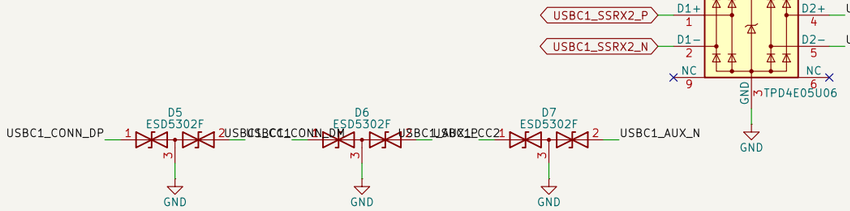
Copied from the Radxa Rock 5B v1.45 schematic, sheet 29 (its one Type-C port).
Port 1 is the power-input port: USBC1_VBUS goes to the power-entry circuit, which is not drawn yet.
Left out: the common-mode choke on the USB 2 pair, the zero-ohm links, and the input-voltage ADC divider.
ESD parts: TPD4E05U06 for the reference's ESD73034D, generic dual TVS for its ESD5302F: confirm equivalents.
The interrupt pull-up goes to 3.3 V to match the GPIO bank (the reference pulls it to 1.8 V).
The processor's OTG ID ball is unused: the port's role comes from the PD controller.

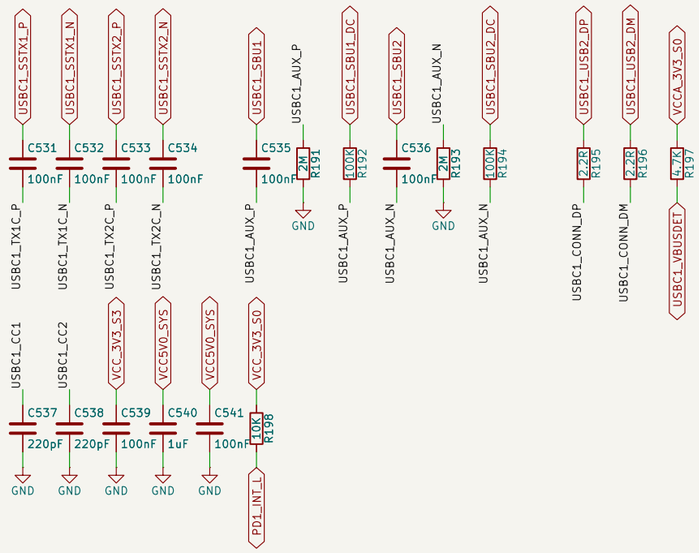












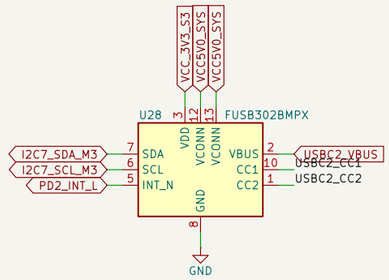
HookProbe

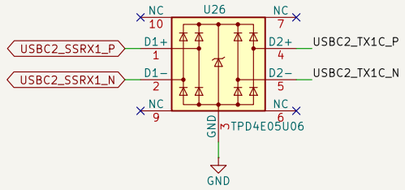
Sheet: /USB-C port 1: USB 3, DisplayPort, power in/
File: usbc1.kicad_sch

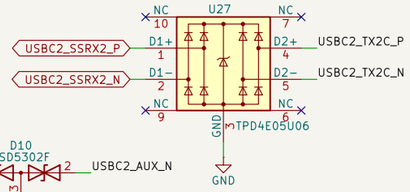
Title: USB-C port 1: USB 3, DisplayPort, power in

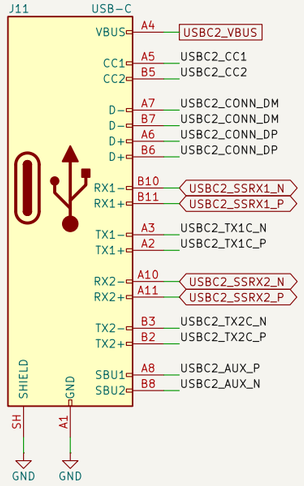
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 19/25

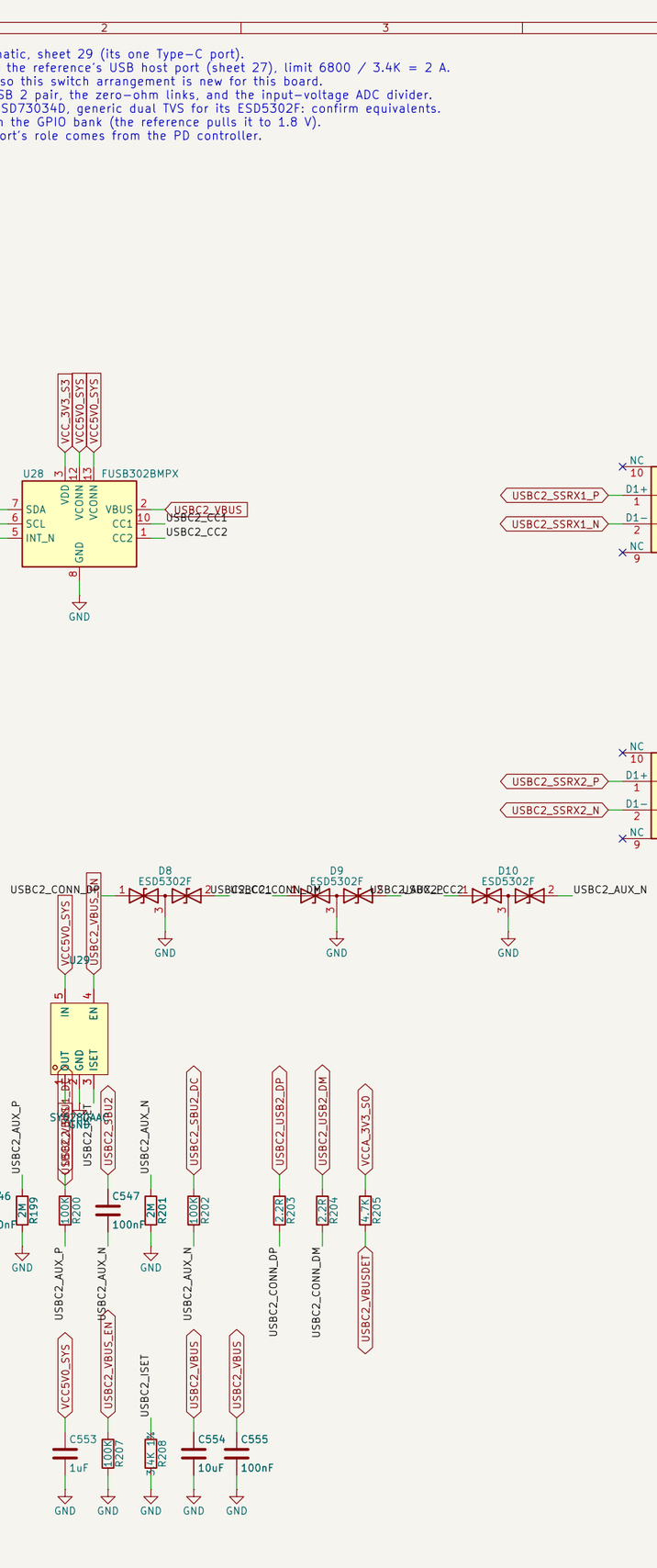
Copied from the Radxa Rock 5B v1.45 schematic, sheet 29 (its one Type-C port).
Port 2 sources 5 V: SY6280AAC switch as on the reference's USB host port (sheet 27), limit 6800 / 3.4K = 2 A.
The reference has no sourcing Type-C port, so this switch arrangement is new for this board.
Left out: the common-mode choke on the USB 2 pair, the zero-ohm links, and the input-voltage ADC divider.
ESD parts: TPD4E05U06 for the reference's ESD730340, generic dual TVS for its ESD5302F: confirm equivalents.
The interrupt pull-up goes to 3.3 V to match the GPIO bank (the reference pulls it to 1.8 V).
The processor's OTG ID ball is unused: the port's role comes from the PD controller.











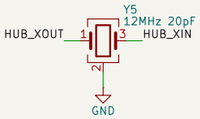
HookProbe

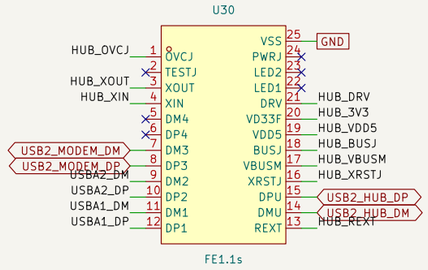
Sheet: /USB-C port 2: USB 3, DisplayPort, power out/
File: usbc2.kicad_sch

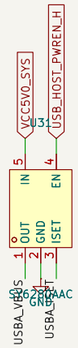
Title: USB-C port 2: USB 3, DisplayPort, power out

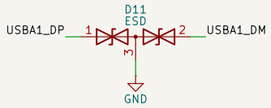
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 20/25

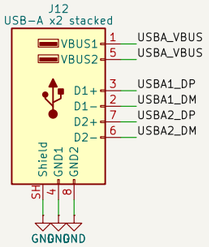
Hub copied from the Radxa Rock 5B v1.45 schematic, sheet 9 (FE1.1s, powered from 3.3 V: VDD5 has only its capacitors, as the hub's datasheet requires). USB-A power switch copied from sheet 27.
Hub ports: 1 and 2 = USB-A, 3 = USB 2 side of the 5G modem slot, 4 = spare.
The USB-A ports are USB 2.0 only. The processor's one spare USB 3 port goes straight to the modem slot; USB 3 on the USB-A ports would need a USB 3 hub chip, for which the reference has no circuit.
One switch feeds both USB-A ports, limit 6800 / 3.4K = 2 A (the reference sets 1 A).
Left out: the common-mode chokes on the USB-A data pairs. ESD parts are generic substitutes.

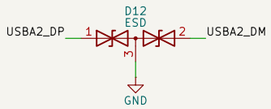


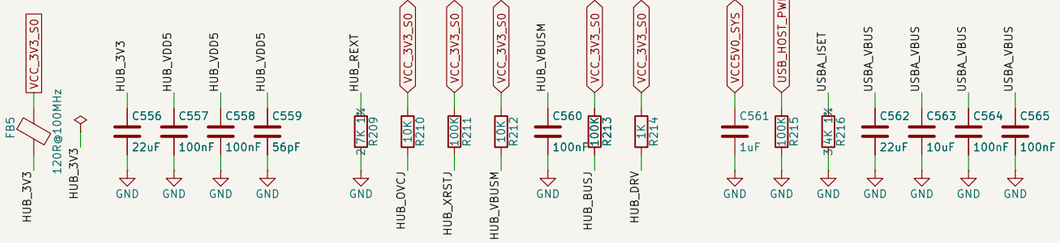


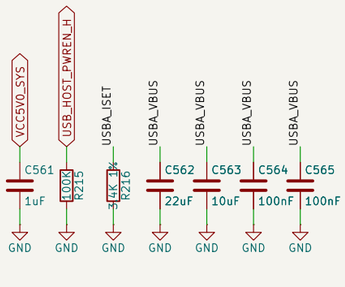












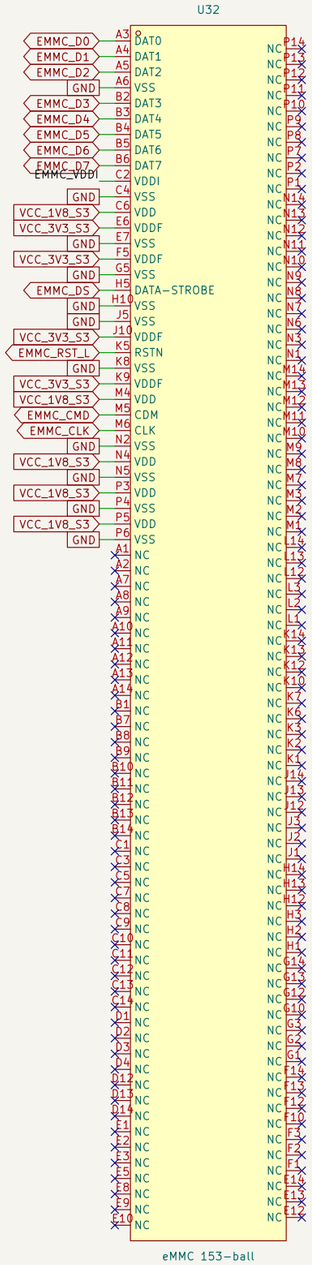
HookProbe

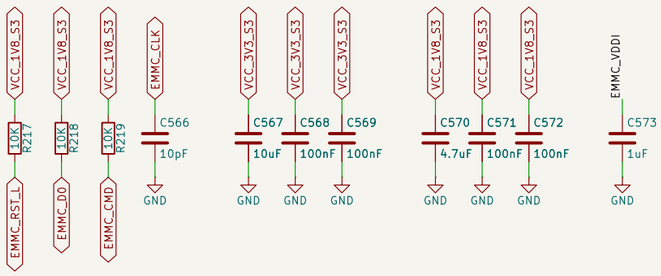
Sheet: /USB 2.0 hub and USB-A ports/
File: usb_hub.kicad_sch

Title: USB 2.0 hub and USB-A ports

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 21/25

The reference has a socket for an eMMC module (sheet 25); this board has the chip soldered down.
From the reference: 10K pull-ups on reset, data 0 and command to VCC_1V8_S3, 10 pF on the clock,
the 3.3 V flash supply from VCC_3V3_S3 and the 1.8 V interface supply from VCC_1V8_S3.
Added for a soldered chip (these sit inside the module on the reference): the 1 uF capacitor on VDDI
and 100 nF capacitors at the supply balls. Check both against the chosen chip's datasheet.
Symbol: Samsung KLMBG2JETD-B041 (32 GB). Any JEDEC 153-ball eMMC 5.1 chip fits the same footprint.





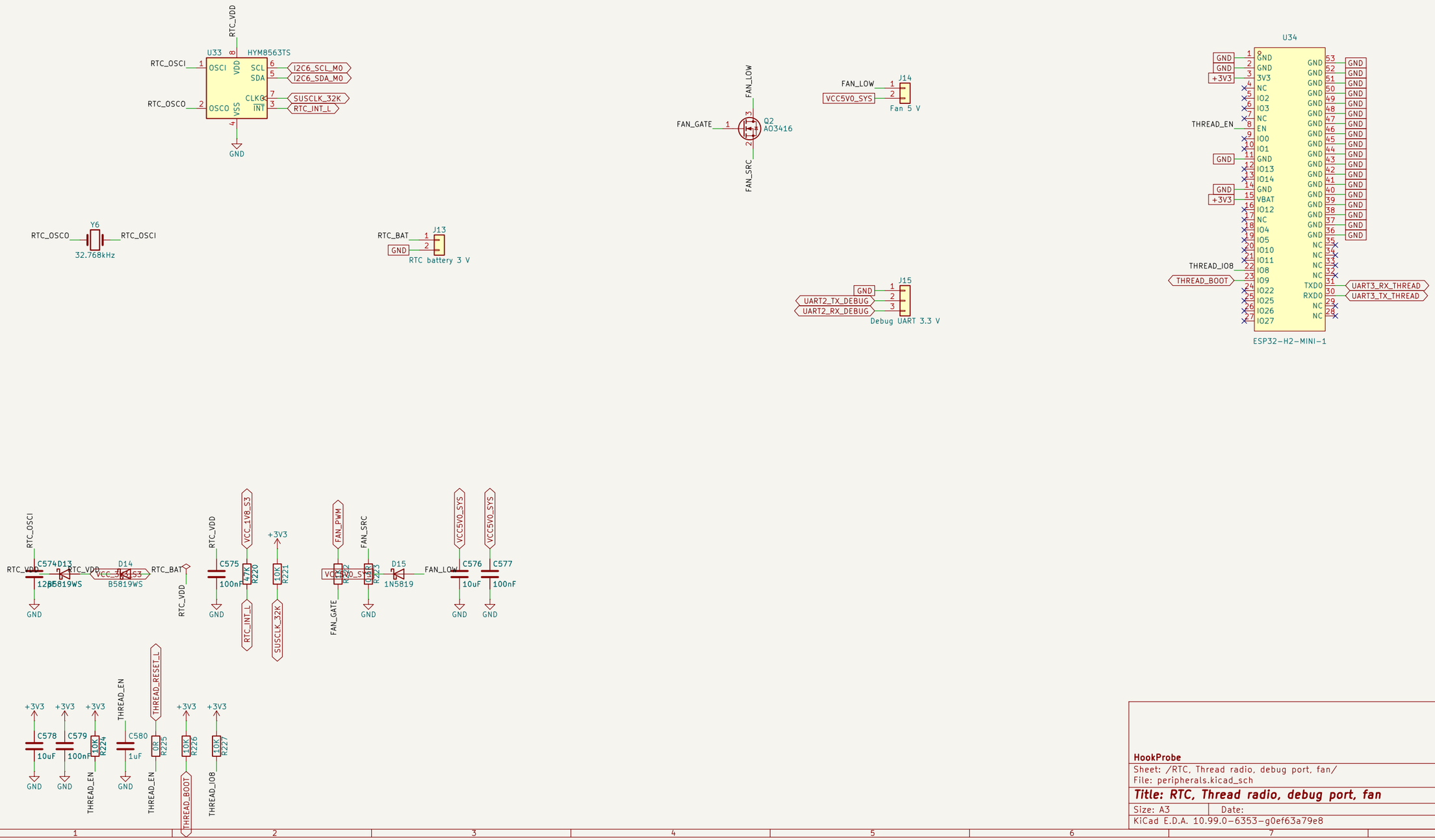
HookProbe

Sheet: /eMMC storage/
File: emmc.kicad_sch

Title: eMMC storage

Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 22/25

RTC: copied from the Radxa Rock 5B v1.45 schematic, sheet 5 (HYM8563, pin-compatible with PCF8563).
  Its 32 kHz output also feeds the Wi-Fi slot; the pull-up goes to +3V3 here (1.8 V on the reference).
Fan: copied from sheet 20 (low-side switch on GPIO0_C0 with a flyback diode).
Thread / Matter radio: ESP32-H2-MINI-1 on UART3. Not from the reference: the enable RC, strap pull-ups
  and pin use follow the module datasheet (vendor/lcsc_C18203134). LCSC showed this module out of stock.
Debug port: UART2 at 3.3 V on a three-pin header.
The TPM is on its own sheet.



HookProbe

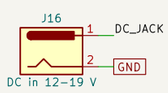
Sheet: /RTC, Thread radio, debug port, fan/
File: peripherals.kicad_sch

Title: RTC, Thread radio, debug port, fan

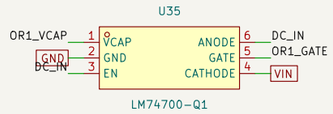
Size: A3	Date:	**Rev: A0**
KiCad E.D.A. 10.99.0-6353-g0ef63a79e8		Id: 23/25

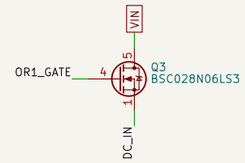
NEW FOR THIS BOARD, not copied from the reference (which has a single USB-C input wired straight to its converters).
Each source reaches VIN through an ideal diode: LM74700-Q1 driving an N-channel MOSFET, wired as its datasheet's
typical application (source = ANODE = input, drain = CATHODE = VIN, EN tied to ANODE, 100 nF from VCAP to ANODE).
This lets the DC jack and USB-C be plugged in together, and blocks a reversed DC plug.
LIMITS TO RESPECT: the converters' absolute maximum input is 26 V. Use 20 V or less. The TVS (SMBJ20A) only limits
short spikes; it clamps well above 26 V at high current, so it does not make a 24 V supply safe.
Still to choose and check: the MOSFETs (40 V, a few milliohms), the fuse rating and the jack part number;
the inrush into the converters' input capacitors when USB-C power is applied; an over-voltage cut-off for field use.

DC jack

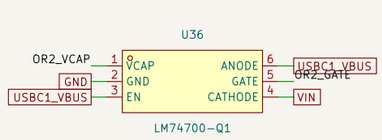


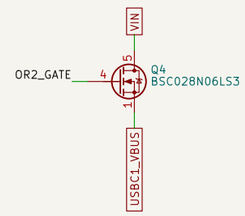


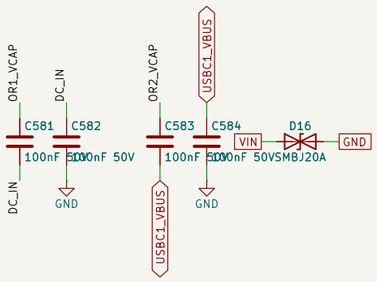




USB-C port 1







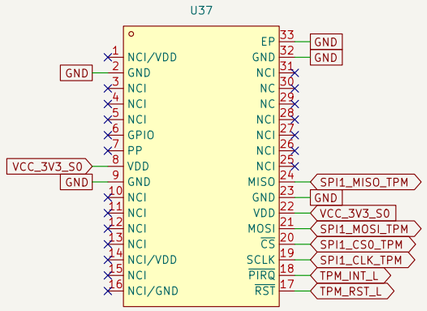
HookProbe

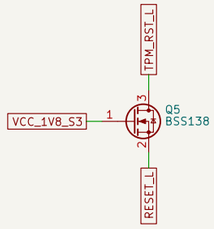
Sheet: /Power entry: DC jack and USB-C input/
File: power_entry.kicad_sch

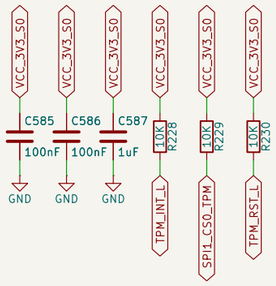
Title: Power entry: DC jack and USB-C input

Size: A3	Date:	**Rev: A0**
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NEW FOR THIS BOARD. Infineon SLB 9670 (the SLB 9672 was not found at LCSC), wired as the typical schematic in its datasheet (vendor/lcsc_C539131): SPI, 100 nF at each VDD pin plus 1 uF, GPIO and PP left open.
Runs from VCC_3V3_S0, the same supply as the processor's GPIO bank 3 it talks to.
TPM reset follows the platform reset (RESET_L, 1.8 V) through a one-transistor level shifter, so software cannot reset the TPM on its own: this is what measured boot relies on.
The interrupt pin is open-drain with no internal pull-up, hence the 10K.







HookProbe

Sheet: /TPM 2.0 security chip/
File: tpm.kicad_sch

Title: TPM 2.0 security chip

Size: A3	Date:	**Rev: A0**
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